

Operator-Valued Matrices with Free or Exchangeable Entries

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ABSTRACT. We study matrices whose entries are free or exchangeable noncommutative elements in some tracial W^* -probability space. More precisely, we consider operator-valued Wigner and Wishart matrices and give for the associated Cauchy transforms explicit rates of convergence to operator-valued semicircular elements over some subalgebra. Applications to block random matrices with a Wigner or circulant structure and in independent or correlated blocks are also given. Our approach relies on a noncommutative extension of the Lindeberg method and Gaussian interpolation techniques.

1. INTRODUCTION

One of the first and fundamental results in random matrix theory is that the empirical spectral distribution of a Wigner matrix whose entries are i.i.d. random variables converges in distribution to the semicircular law. In the noncommutative setting, Voiculescu introduced semicircular systems in his fundamental paper [37] and showed that $N \times N$ matrices in *free* circular/seimicircular operators form a semicircular system for any $N \geq 1$. The distributions of matrices with free R -diagonal elements and creation/annihilation operators as entries were then studied in [22] and [31] respectively. However, for identically distributed free entries, one need to take the limit as $N \rightarrow \infty$ to give a nice explicit description of the matrix distribution [30] and obtain the free analogue of Wigner's theorem. Matrices with noncommutative entries were also studied in [35, 5, 9, 18]. In this paper, we prove that the distribution of Wigner/Wishart matrices in free entries with possibly different variances is close to that of an operator-valued semicircular element over some subalgebra. More precisely, we provide explicit quantitative estimates for the associated Cauchy transforms which can be passed in some cases to the Kolmogorov distance.

Another interesting topic in random matrix theory is the asymptotic spectral behavior of random block matrices [11, 29, 25]. Such matrices are special cases of operator-valued matrices for they can be seen as matrices with entries in some noncommutative algebra. In particular, i.i.d. random matrices are exchangeable in the noncommutative space of random matrices and fit nicely in the framework of matrices with exchangeable entries in some noncommutative probability space.

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The main contribution of this paper is an extension of Voiculescu's result to operator-valued matrices with exchangeable entries. Indeed, we relax the freeness hypothesis and consider the case where the matrix entries consist of a *finite* family of exchangeable elements in some finite-dimensional W^* -probability space $(\mathcal{M}, \tau)$; i.e. their distribution with respect to τ is invariant under permutations. Again, we prove that the distribution of Wigner/Wishart matrices is close to that of an operator-valued semicircular element over some subalgebra and give precise quantitative estimates for the associated Cauchy transforms.

As applications, we consider random matrices with independent blocks, as in [11], and random matrices in which the blocks are themselves correlated but have independent i.i.d. entries, as in [29, 25]. In each case, we give quantitative estimates on the associated Cauchy transforms which might lead to convergence in Kolmogorov distance.

A main ingredient in our proof concerns an independence structure hidden behind exchangeability. In fact, we prove in Section 3.2 that sums of exchangeable operators are close in distribution to the expectation of sums of independent averaged Gaussian operators. The distribution of matrices in random operators, is then shown to be close to that of an operator-valued semicircular element. The appearance of independent Gaussian operators might be surprising or seem unnatural but it could be explained by the fact that invariance under permutations is a commutative concept.

Our approximation technique relies on the Lindeberg method which we extend to Cauchy transforms in the noncommutative setting. This method is also known in probability theory by the replacement trick and has been first introduced by Lindeberg [17] to give an alternative elegant and illustrative proof of the CLT of sums of independent random variables. It has later become widely applied in various problems and specially for establishing explicit rates of convergence. Contributions and refinements were later carried on this method to extend it to more general functions than sums and also for dependent variables. In the context of random matrices, it was first extended by Chatterjee [6] to study matrices with exchangeable entries, by Tao and Vu [33, 34] to study the universality of local laws, by Banna, Merlevède and Peligrad [2, 3, 1] to study matrices with correlated entries, ... etc. In the noncommutative setting, this method was employed to polynomials in [12] to generalize Voiculescu's free CLT and in [8] to prove an invariance principle for multilinear homogeneous sums in free elements. In both cases, it was used in the free case and to give explicit estimates on moments. In this paper, we extend this method to Cauchy transforms and refine it so that we still get good quantitative estimates also in the exchangeable case.

The paper is organized as follows: we recall in Section 2 some basic definitions and tools from free probability theory. In Section 3, the Lindeberg method is extended to the noncommutative setting and then applied to general sums in free or exchangeable elements. Being of independent interest, the proof of the main step of the exchangeable case is given separately in Section 7. These results are then applied in Section 4 to study the distribution of operator-valued Wigner matrices with free or exchangeable coefficients which is in turn extended in Section 5 to cover operator-valued Wishart matrices. Finally applications to block random matrices with a Wigner or circulant structure and in independent or correlated blocks are given in Section 6.

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2. PRELIMINARIES

In this section, we fix the notation and review some concepts and tools from the scalar- and operator-valued free probability theory.

Noncommutative probability spaces. We refer to a pair $(\mathcal{M}, \tau)$, consisting of a von Neumann algebra $\mathcal{M}$ and a faithful normal tracial state $\tau : \mathcal{M} \rightarrow \mathbb{C}$, as a *tracial W^* -probability space*. The *operator norm* is denoted by $\|\cdot\|_{L^\infty(\mathcal{M}, \tau)}$, or by $\|\cdot\|_\infty$ when the context is clear. The *L^p -norms* are defined, for all $p \geq 1$ and all $a \in \mathcal{M}$, by

$$\|a\|_{L^p(\mathcal{M}, \tau)} = [\tau|a|^p]^{1/p},$$

and can be denoted by $\|\cdot\|_{L^p}$ when the context is sufficiently clear.

If $(\mathcal{M}, \tau)$ and $(\mathcal{N}, \varphi)$ are two tracial W^* -probability spaces, the tracial W^* -probability space $(\mathcal{M} \otimes \mathcal{N}, \tau \otimes \varphi)$ is the tensor product $\mathcal{M} \otimes \mathcal{N}$ of von Neumann algebra endowed with the unique faithful normal tracial state $\tau \otimes \varphi$ such that, for all $a \in \mathcal{M}$ and $b \in \mathcal{N}$, $\tau \otimes \varphi(a \otimes b) = \tau(a)\varphi(b)$. For all $a \in \mathcal{M}$ and $b \in \mathcal{N}$ and $p \in [1, \infty]$ we have

$$\|a \otimes b\|_{L^p(\mathcal{M} \otimes \mathcal{N}, \tau \otimes \varphi)} = \|a\|_{L^p(\mathcal{M}, \tau)} \cdot \|b\|_{L^p(\mathcal{N}, \varphi)}.$$

Finally, for all $a \in \mathcal{M}$ and $b \in \mathcal{N}$, we let $a \otimes^* b := a \otimes b + a^* \otimes b^*$.

Exchangeability. Elements $x_1, \dots, x_n \in \mathcal{M}$ are said to be *exchangeable* with respect to τ if, for any polynomial P in $2n$ noncommutative variables, and any permutation $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, we have

$$\tau[P(x_1, \dots, x_n, x_1^*, \dots, x_n^*)] = \tau[P(x_{\sigma(1)}, \dots, x_{\sigma(n)}, x_{\sigma(1)}^*, \dots, x_{\sigma(n)}^*)].$$

An infinite sequence $(x_n)_{n \geq 1}$ in $\mathcal{M}$ is called *exchangeable* with respect to τ if, for any $n \geq 1$, $x_1, \dots, x_n \in \mathcal{M}$ are exchangeable. The tail subalgebra $\mathcal{M}_{tail}$ is defined as $\bigcap_{n \geq 1} \mathcal{B}_n$, where $\mathcal{B}_n$ is the von Neumann subalgebra generated by $(x_k)_{k \geq n}$.

Cauchy Transform. For all element $a \in \mathcal{M}$ with negative imaginary part, we denote by G_a the *resolvent* $G_a(z) = (z1_{\mathcal{M}} - a)^{-1}$, which is defined on the upper half-plane $\mathbb{C}^+ = \{x + iy \in \mathbb{C} : y > 0\}$. If a is self-adjoint, the resolvent captures the spectral properties of a : for all $z \in \mathbb{C}^+$,

$$\tau[G_a(z)] = \int_{\mathbb{R}} \frac{1}{z - x} d\mu(x),$$

where μ is the *spectral measure* of a , i.e. the unique probability measure on $\mathbb{R}$ such that the moments of a are the same as the moments of the probability measure μ . In other words, $\tau \circ G_a$ is the Cauchy transform of the spectral measure of a . As an immediate consequence, the convergence, as n goes to ∞ , of $\tau[G_{a_n}(z)]$ to the Cauchy transform of a measure ν implies the weak convergence of the spectral measure of a_n to ν .

Operator-valued semicircular. If $\mathcal{B}$ is a von Neumann subalgebra of $\mathcal{M}$, there exists a unique linear map $\tau[\cdot|\mathcal{B}] : \mathcal{M} \rightarrow \mathcal{B}$, called the *conditional expectation* onto $\mathcal{B}$, such that for all $a \in \mathcal{M}$ and $b_1, b_2 \in \mathcal{B}$

$$\tau[b_1 a b_2 | \mathcal{B}] = b_1 \tau[a | \mathcal{B}] b_2 \quad \text{and} \quad \tau[\tau[a | \mathcal{B}]] = \tau[a].$$

Let $\eta : \mathcal{B} \rightarrow \mathcal{B}$ be a completely positive linear mapping. Then, the *operator-valued Cauchy transform of an operator-valued semicircular element* with covariance mapping η is the unique analytic map $g : \mathbb{C}^+ \rightarrow \mathcal{B}$ with negative imaginary part such that for all $z \in \mathbb{C}^+$,

$$zg(z) = 1_{\mathcal{A}} + \eta(g(z)) \cdot g(z),$$

and $\lim_{|z| \rightarrow \infty} zg(z) = 1$. The name of g comes from the fact that g can be represented as

$$g(z) = \tau[G_s(z) | \mathcal{B}]$$

where s is an *operator-valued semicircular element with variance* η . We refer to [20, Chapter 9] for the rigorous definition of an operator-valued semicircular element and more details on operator-valued Cauchy transforms. In practice, $g(z)$ can be computed as the limit of iterates $g_n = G_{\eta(g_{n-1})}(z)$ for any initial point g_0 with negative imaginary part [20, Section 10.4.2].

Let $t > 0$. A *semicircular element with variance* t (or with covariance mapping $\eta : \mathbb{C} \ni z \rightarrow tz \in \mathbb{C}$) is just a self-adjoint element s whose spectral measure is the semicircular measure

$$\frac{1}{2\pi t} \sqrt{4t - x^2} \cdot 1_{[-2\sqrt{t}, 2\sqrt{t}]}(x) dx.$$

Freeness and freeness with amalgamation. We say that unital subalgebras $(\mathcal{A}_i)_{i \in I}$ of $\mathcal{M}$ are *free* if whenever we have $m \geq 2$ and $a_1, \dots, a_m \in \mathcal{M}$ such that $\tau[a_j] = 0$ for $j = 1, \dots, m$, $a_j \in \mathcal{A}_{i(j)}$ with $i(j) \in I$ and $i(1) \neq i(2), i(2) \neq i(3), \dots, i(m-1) \neq i(m)$, then

$$\tau[a_1 \cdots a_m] = 0.$$

Elements in $\mathcal{M}$ are called *freely independent* or *free* if the $*$ -algebras generated by them are free. Similarly we define *freeness with amalgamation* over the operator-valued probability space $\mathcal{B} \subset \mathcal{A}_i$ for all $i \in I$. The subalgebras $(\mathcal{A}_i)_{i \in I}$ of $\mathcal{M}$ are *free with amalgamation over* $\mathcal{B}$ (or *free with amalgamation* with respect to the conditional expectation $\tau[\cdot|\mathcal{B}]$), if whenever we have $m \geq 2$ and $a_1, \dots, a_m \in \mathcal{M}$ such that $\tau[a_j|\mathcal{B}] = 0$ for $j = 1, \dots, m$, $a_j \in \mathcal{A}_{i(j)}$ with $i(j) \in I$ and $i(1) \neq i(2), i(2) \neq i(3), \dots, i(m-1) \neq i(m)$, then

$$\tau[a_1 \cdots a_m | \mathcal{B}] = 0.$$

Elements in $\mathcal{M}$ are called *free with amalgamation over* $\mathcal{B}$ if the algebras generated by $\mathcal{B}$ and the elements are also so.

3. THE NONCOMMUTATIVE LINDBERG METHOD

Our main results rely on the Lindeberg method which is a well-known approximation technique in classical probability theory. The aim of this section is to extend this method to the noncommutative setting of a tracial W^* -probability space $(\mathcal{M}, \tau)$. More precisely, we shall compare Cauchy transforms of elements of the form $\sum_i x_i$ with those of some other elements $\sum_i y_i$ in $\mathcal{M}$. The main idea behind this technique is to write the difference of the resolvents as a telescoping sum, replace the x_i 's by the y_i 's one at a time and finally give quantitative error bounds. This swapping process is known as the Lindeberg *replacement trick* and allows such approximations under mild moment conditions.

Theorem 3.1. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space. Let $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ be n -tuples of self-adjoint elements in $\mathcal{M}$. Setting for any $i \in \{1, \dots, n\}$,*

$$\mathbf{z}_i = x_1 + \dots + x_i + y_{i+1} + \dots + y_n$$

and

$$\mathbf{z}_i^0 = x_1 + \dots + x_{i-1} + y_{i+1} + \dots + y_n$$

then, for any $z \in \mathbb{C}^+$, we have

$$\left| \tau[G_{\mathbf{z}_n}(z)] - \tau[G_{\mathbf{z}_0}(z)] \right| \leq \sum_{i=1}^n (|P_i| + |Q_i| + |R_i|),$$

where, for any $i \in \{1, \dots, n\}$,

$$P_i = \tau[G_{\mathbf{z}_i^0}(z)(x_i - y_i)G_{\mathbf{z}_i^0}(z)],$$

$$Q_i = \tau[G_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)] - \tau[G_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)],$$

and

$$R_i = \tau[G_{\mathbf{z}_i}(z)(x_iG_{\mathbf{z}_i^0}(z))^3] - \tau[G_{\mathbf{z}_{i-1}}(z)(y_iG_{\mathbf{z}_i^0}(z))^3].$$

Moreover, if $x_1, \dots, x_n, y_1, \dots, y_n$ are random self-adjoint elements in $\mathcal{M}$, we have

$$\left| \mathbb{E}[\tau[G_{\mathbf{z}_n}(z)] - \tau[G_{\mathbf{z}_0}(z)]] \right| \leq \sum_{i=1}^n (|\mathbb{E}[P_i]| + |\mathbb{E}[Q_i]| + |\mathbb{E}[R_i]|).$$

Remark 3.2. The above theorem can be easily extended to more general differentiable functions $f(x_1, \dots, x_n)$ in n noncommuting variables and is not restricted to Cauchy transforms of partial sums. In that case, the terms of the telescoping sum are developed using Taylor expansions and then the estimates are obtained in terms of noncommutative partial derivatives. However, the restriction we made was essential for our purposes and in particular for dealing with exchangeable variables in Section 3.2.

Proof of Theorem 3.1. For any $z \in \mathbb{C}^+$, we write the difference of the resolvents as a telescoping sum as follows:

$$G_{\mathbf{z}_n}(z) - G_{\mathbf{z}_0}(z) = \sum_{i=1}^n G_{\mathbf{z}_i}(z) - G_{\mathbf{z}_{i-1}}(z) = \sum_{i=1}^n (G_{\mathbf{z}_i}(z) - G_{\mathbf{z}_i^0}(z)) - \sum_{i=1}^n (G_{\mathbf{z}_{i-1}}(z) - G_{\mathbf{z}_i^0}(z)).$$

To pursue the proof, we shall use the following expansion:

Lemma 3.3. *Let a and b be invertible in $\mathcal{M}$, then*

$$a^{-1} = b^{-1} + a^{-1}(b - a)b^{-1}$$

and more generally, for each $m \in \mathbb{N}_0$, the Taylor formula (with remainder term) holds:

$$a^{-1} = \sum_{k=0}^m b^{-1}[(b - a)b^{-1}]^k + a^{-1}[(b - a)b^{-1}]^{m+1}.$$

Applying Lemma 3.3 with $m = 3$, gives for any $i \in \{1, \dots, n\}$,

$$\begin{aligned} G_{\mathbf{z}_i}(z) - G_{\mathbf{z}_i^0}(z) &= G_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z) + G_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z) \\ &\quad + G_{\mathbf{z}_i}(z)x_iG_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z), \end{aligned}$$

and analogously,

$$\begin{aligned} G_{\mathbf{z}_{i-1}}(z) - G_{\mathbf{z}_i^0}(z) &= G_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z) + G_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z) \\ &\quad + G_{\mathbf{z}_{i-1}}(z)y_iG_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z). \end{aligned}$$

The assertion follows by taking their difference and applying τ or $\mathbb{E} \circ \tau$ in the random variant. $\square$

3.1. Freeness. We address in this section sums of freely independent elements in $(\mathcal{M}, \tau)$. We shall see how a simple application of the noncommutative Lindeberg method in Theorem 3.1 gives quantitative estimates to the difference of Cauchy transforms under the mere condition of matching conditional expectations of first and second orders.

Theorem 3.4. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space and $\mathcal{B}$ be a von Neumann subalgebra of $\mathcal{M}$. Let $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ be two n -tuples of self-adjoint elements in $\mathcal{M}$ which are free with amalgamation over $\mathcal{B}$ and such that $\tau[x_i|\mathcal{B}] = \tau[y_i|\mathcal{B}] = 0$ and $\tau[x_i b x_i|\mathcal{B}] = \tau[y_i b y_i|\mathcal{B}]$ for any $b \in \mathcal{B}$ and $1 \leq i \leq n$. Setting*

$$\mathbf{x} = \sum_{i=1}^n x_i \text{ and } \mathbf{y} = \sum_{i=1}^n y_i$$

then, for any $z \in \mathbb{C}^+$, we have

$$|\tau[G_{\mathbf{x}}(z)] - \tau[G_{\mathbf{y}}(z)]| \leq \frac{K_{\infty} K_2^2}{\text{Im}(z)^4} n,$$

where $K_{\infty} = \max_i (\|x_i\|_{\infty} + \|y_i\|_{\infty})$ and $K_2 = \max_i (\|x_i\|_{L^2} + \|y_i\|_{L^2})$.

Proof. Let us assume without loss of generality that $(x_1, \dots, x_n, y_1, \dots, y_n)$ are free. The proof is a direct application of the Lindeberg method in Theorem 3.1. Note that for any $i = 1, \dots, n$, x_i and y_i are centered and free from $\mathbf{z}_i^0$ with amalgamation over $\mathcal{B}$ which implies that

$$\tau[G_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)] = \tau[G_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)] = 0$$

and thus $P_i = 0$. For the same reasons

$$\begin{aligned} \tau[G_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)] &= \tau[G_{\mathbf{z}_i^0}(z)\tau[x_iG_{\mathbf{z}_i^0}(z)x_i|\mathcal{B}]G_{\mathbf{z}_i^0}(z)] \\ &= \tau[G_{\mathbf{z}_i^0}(z)\tau[y_iG_{\mathbf{z}_i^0}(z)y_i|\mathcal{B}]G_{\mathbf{z}_i^0}(z)] \\ &= \tau[G_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)] \end{aligned}$$

implying that $Q_i = 0$. Considering the third order term R_i , we compute

$$\begin{aligned} \tau \left[G_{\mathbf{z}_i}(z) (x_i G_{\mathbf{z}_i^0}(z))^3 \right] &\leq \|G_{\mathbf{z}_i}(z) (x_i G_{\mathbf{z}_i^0}(z))^3\|_{L^1} \\ &\leq \|G_{\mathbf{z}_i}(z)\|_\infty \|G_{\mathbf{z}_i^0}(z)\|_\infty^3 \|x_i\|_\infty \|x_i\|_{L^2}^2 \\ &\leq \frac{1}{\operatorname{Im}(z)^4} \|x_i\|_\infty \|x_i\|_{L^2}^2, \end{aligned}$$

and similarly, we control the second term in R_i and get that $R_i \leq \frac{K_\infty K_2^2}{\operatorname{Im}(z)^4}$. We finally conclude by summing over i . $\square$

3.2. Exchangeable variables. In this section, we drop the free independence hypothesis and treat instead exchangeable variables in $(\mathcal{M}, \tau)$. We shall see how we can approximate sums of exchangeable operators with independent Gaussian operators and give again quantitative estimates to such approximations in terms of Cauchy transforms.

Theorem 3.5. *Let $(\mathcal{M}, \tau)$ and $(\mathcal{N}, \varphi)$ be two tracial W^* -probability spaces. Let $(x_1, \dots, x_n)$ be an n -tuple of exchangeable elements in $\mathcal{M}$. Consider a family $(N_{i,k})_{1 \leq i, k \leq n}$ of independent standard Gaussian random variables and let $(y_1, \dots, y_n)$ be the n -tuple of random elements in $\mathcal{M}$ given by*

$$y_i = \frac{1}{\sqrt{n}} \sum_{k=1}^n N_{i,k} \left(x_k - \frac{1}{n} \sum_{j=1}^n x_j \right).$$

Let $(a_1, \dots, a_n)$ be an n -tuple of elements in $\mathcal{N}$ and set

$$\mathbf{x} = \sum_{i=1}^n x_i \otimes a_i + x_i^* \otimes a_i^* \quad \text{and} \quad \mathbf{y} = \sum_{i=1}^n y_i \otimes a_i + y_i^* \otimes a_i^*.$$

Then, for any $z \in \mathbb{C}^+$,

$$\begin{aligned} & \left| \tau \otimes \varphi(G_{\mathbf{x}}(z)) - \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{y}}(z))] \right| \\ & \leq C \cdot \left(\frac{K_1}{\operatorname{Im}(z)^2} \|x_1\|_\infty + \frac{K_2^2}{\operatorname{Im}(z)^3} \|x_1\|_\infty^2 \sqrt{n} + \frac{K_\infty K_2^2}{\operatorname{Im}(z)^4} \|x_1\|_\infty^3 n + \frac{K_\infty^2 K_2^2}{\operatorname{Im}(z)^5} \|x_1\|_\infty^4 n \right), \end{aligned}$$

where $K_\infty = \max_i \|a_i\|_\infty$, $K_2 = \max_i \|a_i\|_{L^2}$, $K_1 = \|\sum_{i=1}^n a_i\|_{L^1}$ and C is a universal constant.

Passing directly from sums of exchangeable elements to semicirculars was not possible under our mild conditions and even under stronger ones, like quantum exchangeability [38, page 108], and this intermediate approximation with random operators was crucial to our purposes.

Being long and technical, the proof of Theorem 3.5 is postponed to Section 7. The results we have presented so far cover quite general sums in tracial W^* -probability spaces. In the following, we specify our object of interest and study precisely operator-valued matrices. We start by considering Wigner matrices and then extend the results obtained to cover Wishart matrices.

4. OPERATOR-VALUED WIGNER MATRICES

This section is devoted for the study of Wigner matrices with operator-valued entries. Let $(\mathcal{A}, \varphi)$ be a tracial W^* -probability space and $\mathbf{x} = (x_{ij})_{1 \leq j \leq i \leq N}$ be a finite family of elements in $\mathcal{A}$. An operator-valued Wigner matrix $A(\mathbf{x})$ of dimension $N \times N$ is the matrix given by

$$[A(\mathbf{x})]_{ij} = \frac{1}{\sqrt{N}} \begin{cases} x_{ij} & \text{if } j < i \\ x_{ji}^* & \text{if } j > i \\ (x_{ii} + x_{ii}^*)/2 & \text{if } i = j. \end{cases}$$

To study its behavior, we consider the Cauchy transform and apply the results in Sections 3.1 and 3.2. With this aim, we set $n = N(N+1)/2$ and note that the associated linear map $A : \mathcal{A}^n \rightarrow M_N(\mathcal{A})$ can be written as follows:

$$(4.1) \quad A(\mathbf{x}) = \sum_{1 \leq j \leq i \leq N} (x_{ij} \otimes a_{ij} + x_{ij}^* \otimes a_{ij}^*) =: \sum_{1 \leq j \leq i \leq N} x_{ij} \otimes^* a_{ij},$$

where $a_{ii} = \frac{1}{2\sqrt{N}}E_{ii}$ and $a_{ij} = \frac{1}{\sqrt{N}}E_{ij}$ for $j < i$ and $(E_{ij})_{1 \leq i, j \leq N}$ are the canonical $N \times N$ matrices.

4.1. Free entries. In this section, we go back to Voiculescu's result on obtaining semi-circular distributions from matrices with free circular/semicircular entries. More precisely, Voiculescu proved in [37] that, for any $N \geq 1$, an $N \times N$ Wigner matrix with free circular off-diagonal entries and free semicircular diagonal entries, is itself semicircular. This result was later generalized in [22, Corollary 3.3] to matrices with free self-adjoint elements on the diagonal and free R -diagonal elements (introduced in [23]) otherwise. The authors give, for any $N \geq 1$, the distribution of the matrix in terms of its R -transform under some summation conditions on the free cumulants of the entries of each row. In general, one need to take the limit as $N \rightarrow \infty$ to give an explicit description of the *limiting* distribution of the matrix. For the case of free identically distributed entries, a semicircular element appears in the limit [30, 18].

In this section, we show that a Wigner matrix with free entries, that can possess different variances, is close in distribution to a matrix in free circular/semicircular elements with the same variance structure. The latter is an operator-valued semicircular element over the subalgebra of scalar diagonal matrices. Moreover, we give explicit quantitative estimates for the associated Cauchy transforms on the upper complex half-plane.

Theorem 4.1. *Let $(\mathcal{A}, \varphi)$ be a tracial W^* -probability spaces. Let $\mathbf{x} = (x_{ij})_{1 \leq j \leq i \leq N}$ and $\mathbf{y} = (y_{ij})_{1 \leq j \leq i \leq N}$ be two families of free variables in $\mathcal{A}$. Assume that*

- $x_{ii}^* = x_{ii}$ for all i ,
- y_{ii} is a semicircular element for all i and y_{ij} is a circular element for all $j < i$,
- $\varphi(x_{ij}) = \varphi(y_{ij}) = 0$ and $\varphi(x_{ij}^* x_{ij}) = \varphi(y_{ij}^* y_{ij})$ for all $j \leq i$,
- $\varphi(x_{ij}^2) = 0$ for all $j < i$.

Then for any $z \in \mathbb{C}^+$,

$$|(\varphi \otimes \text{tr}_N)[G_{A(\mathbf{x})}(z)] - (\varphi \otimes \text{tr}_N)[G_{A(\mathbf{y})}(z)]| = \mathcal{O}\left(\frac{1}{\sqrt{N}}\right),$$

where the constant in $\mathcal{O}(1/\sqrt{N})$ depends on z and $\max_{1 \leq j \leq i \leq N} \|x_{ij}\|_\infty$.

$A(\mathbf{y})$ is an operator-valued semicircular element over the subalgebra $\mathcal{D} \subset M_N(\mathbb{C})$ of diagonal matrices whose variance is given by the completely positive map $\eta : \mathcal{D} \rightarrow \mathcal{D}$ defined by

$$[\eta(b)]_{ij} = \frac{1}{N} \sum_{k=1}^N \varphi(x_{ik}x_{jk}^*)b_{kk}, \quad \text{for } b = [b_{k\ell}]_{k,\ell=1}^N$$

As a consequence of [21, Theorem 3.3], $A(\mathbf{y})$ is itself a semicircular element in $\mathcal{A}$ if and only if $\eta(1)$ is a multiple of the identity which is the case only when the sums of variances in each row of the matrix are the same. In particular, if all the entries have the same variance, i.e. $\varphi(x_{ij}^*x_{ij}) = \sigma^2$ for all $j \leq i$, then $A(\mathbf{y})$ has a semicircular distribution. Note that when the cumulative distribution of $A(\mathbf{y})$ is Lipschitz, the estimate on the Cauchy transform gives quantitative estimates on the Kolmogorov distance [32, Theorem A.2]. One can check Chapter 9 in [20] for more details on operator-valued free probability.

Remark 4.2. Another relevant question is when, for any $N \geq 1$, an $N \times N$ operator-valued Wigner matrix has itself the same distribution as its entries. Indeed this case exists and it was shown by Shlyakhtenko [31] that a Wigner matrix in $M_N(\mathcal{A})$ with free creation operators as entries is itself a creation operator in $\mathcal{A}$. Later, Ryan [30, Corollary 29] showed that Shlyakhtenko's example is close to be the only one. Otherwise he showed that the same distribution can be only obtained when taking the limit as $N \rightarrow \infty$.

Proof of Theorem 4.1. The proof of this theorem is a direct application of Theorem 3.4 where we note that in this case $\mathcal{M} = M_N(\mathcal{A})$ with $\tau = \varphi \otimes \text{tr}_N$ as state and $\varphi \otimes \text{id} : M_N(\mathcal{A}) \rightarrow M_N(\mathbb{C})$ as the conditional expectation over $\mathcal{B} := M_N(\mathbb{C}) \subset M_N(\mathcal{A})$. It is easy to see that $\varphi \otimes \text{id}[x_{ij} \otimes^* a_{ij}] = \varphi \otimes \text{id}[y_{ij} \otimes^* a_{ij}] = 0$ and that for any $b \in M_N(\mathbb{C})$

$$\varphi \otimes \text{id} [(x_{ij} \otimes^* a_{ij})b(x_{ij} \otimes^* a_{ij})] = \varphi \otimes \text{id} [(y_{ij} \otimes^* a_{ij})b(x_{ij} \otimes^* y_{ij})].$$

The result follows by finally noting that $K_\infty = \mathcal{O}(1/\sqrt{N})$ and $K_2 = \mathcal{O}(1/N)$. $\square$

4.2. Exchangeable entries. Independent and identically distributed matrices can be seen as exchangeable elements in some tracial W^* -probability space $\mathcal{M}$ and thus many random block matrices fit nicely in the framework of operator-valued matrices with exchangeable entries. This section is devoted for the study of the behavior of such matrices and in particular of Hermitian matrices with Wigner structure. More precisely, we consider a *finite* family of exchangeable operators as entries and prove that the associated Wigner matrix is close in distribution to an operator-valued semicircular element by providing explicit rates of convergence for the Cauchy transforms.

Theorem 4.3. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space with $\mathcal{M}$ being a direct integral of finite dimensional factors of dimension $\leq d$. Let $\mathbf{x} = (x_{ij})_{1 \leq j \leq i \leq N}$ be a family of exchangeable variables in $\mathcal{M}^n$ with $n = N(N+1)/2$. Set $\bar{x}_{ij} = x_{ij} - \mu$ where $\mu = \frac{1}{n} \sum_{1 \leq j \leq i \leq N} x_{ij}$ and consider their variance function*

$$\eta : \mathcal{M} \rightarrow \mathcal{M}, \quad \eta(b) = \frac{1}{n} \sum_{j \leq i} \bar{x}_{ij} b \bar{x}_{ij}.$$

Let $\mathcal{B}$ be the smallest W^ -algebra which is closed under η and consider a centered $\mathcal{B}$ -valued semicircular variable S of variance $E_{\mathcal{B}}(SbS) = \eta(b)$. Letting $A(\mathbf{x}) \in \mathcal{M}_N(\mathcal{M})$ be the*

operator-valued matrix defined in (4.1) then for any z such that $\text{Im}(z) > \|\eta\|^{1/2}$, we have

$$|(\tau \otimes \text{tr}_N)[G_{A(\mathbf{x})}(z)] - \tau[G_S(z)]| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right).$$

More precisely, there exists a universal constant C such that

$$\begin{aligned} & |(\tau \otimes \text{tr}_N)[G_{A(\mathbf{x})}(z)] - \tau[G_S(z)]| \\ & \leq \frac{C}{\sqrt{N}} \left(\frac{\|x_{11}\|_\infty}{\text{Im}(z)^2} + \frac{\|x_{11}\|_\infty^3}{\text{Im}(z)^4} \right) + \frac{C}{N} \left(\frac{\|x_{11}\|_\infty^2}{\text{Im}(z)^3} + \frac{\|x_{11}\|_\infty^4}{\text{Im}(z)^5} + \frac{d}{\text{Im}(z)^2 - \|x_{11}\|_\infty^2} \frac{\|x_{11}\|_\infty^3}{\text{Im}(z)^2} \right). \end{aligned}$$

Remark 4.4. If $\eta(1_{\mathcal{M}}) = \frac{1}{n} \sum_{j \leq i} \hat{x}_{ij}^2 = \sigma 1_{\mathcal{M}}$ then the semicircular element S considered in Theorem 4.3 is simply a centered semicircular variable of variance σ .

The proof of Theorem 4.3 is divided into three parts: first we shall apply the Lindeberg method to approximate the Cauchy transform of $A(\mathbf{x})$ by that of a random Gaussian operator-valued matrix, change of covariance by an interpolation procedure, and then prove that the later is close to the Cauchy transform of the semicircular element S .

The first step will be accomplished in the following Theorem which is an application of the Lindeberg method in the exchangeable setting proved in Section 3.2:

Theorem 4.5. Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space. Let $\mathbf{x} = (x_{ij})_{1 \leq j \leq i \leq N}$ be a family of exchangeable variables in $\mathcal{M}^n$, and let us denote by μ their mean. Consider a family $(N_{ij,kl})_{1 \leq j \leq i \leq N, 1 \leq l \leq k \leq N}$ of n^2 independent Gaussian random variables. We consider the random operators $\mathbf{y} = (y_{ij})_{1 \leq j \leq i \leq N}$ given by

$$y_{ij} = \frac{1}{\sqrt{n}} \sum_{1 \leq l \leq k \leq N} N_{ij,kl} (x_{kl} - \mu).$$

Then for any $z \in \mathbb{C}^+$, there exists a universal constant C such that

$$\begin{aligned} & |(\tau \otimes \text{tr}_N)[G_{A(\mathbf{x})}(z)] - \mathbb{E}[(\tau \otimes \text{tr}_N)[G_{A(\mathbf{y})}(z)]]| \\ & \leq C \cdot \left(\frac{\|x_1\|_\infty}{\text{Im}(z)^2} \frac{1}{\sqrt{N}} + \frac{\|x_1\|_\infty^2}{\text{Im}(z)^3} \frac{1}{N} + \frac{\|x_1\|_\infty^3}{\text{Im}(z)^4} \frac{1}{\sqrt{N}} + \frac{\|x_1\|_\infty^4}{\text{Im}(z)^5} \frac{1}{N} \right). \end{aligned}$$

The second step of the proof is simply a change of covariance done in the following theorem:

Theorem 4.6. Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space. Let $\mathbf{x} = (x_{ij})_{1 \leq j \leq i \leq N}$ be a family of variables in $\mathcal{M}^n$, and let us denote by μ their mean. Consider a family $(N_{ij,kl})_{1 \leq j, i \leq N, 1 \leq l \leq k \leq N}$ of $N^2 \times n$ independent Gaussian random variables. We consider the random operators $\mathbf{y} = (y_{ij})_{1 \leq j \leq i \leq N}$ given by

$$y_{ij} = \frac{1}{\sqrt{n}} \sum_{1 \leq l \leq k \leq N} N_{ij,kl} \bar{x}_{kl},$$

where $\bar{x}_{ij} = x_{ij} - \mu$ and consider the random matrix

$$Y := Y_N = \frac{1}{\sqrt{n}} \sum_{i,j=1}^N \sum_{1 \leq l \leq k \leq N} N_{ij,kl} \bar{x}_{kl} \otimes K_{ij} \in \mathcal{M}_N(\mathcal{M}),$$

where K_{ij} is an orthonormal basis of the Hermitian matrices, in such a way that $\sum_{ij} K_{ij} \otimes K_{ij} = \frac{1}{N} \sum_{ij} E_{ij} \otimes E_{ji}$ (a possible choice is $K_{ij} = \frac{1}{2\sqrt{N}}(E_{ij} + E_{ji})$ if $i \leq j$ and $K_{ij} =$

$\frac{1}{2i\sqrt{N}}(E_{ij} - E_{ji})$ if $i > j$). Letting A be the linear map defined in (4.1) then for any z such that $\text{Im}(z) > \|\eta\|^{1/2}$, we have

$$\left| \mathbb{E}[(\tau \otimes \text{tr}_N)(G_{A(\mathbf{y})}(z))] - \mathbb{E}[(\tau \otimes \text{tr}_N)(G_Y(z))] \right| \leq \frac{8\|x_{11}\|_\infty^2}{N\text{Im}(z)^3}.$$

The above theorem shows the change of covariance doesn't affect the asymptotic behavior of the Gaussian operator-valued matrix $A(\mathbf{y})$ and that one can pass the analysis to Y instead. This is achieved via a Gaussian interpolation technique stated in the following lemma:

Lemma 4.7. *Let $(\mathcal{N}, \varphi)$ be a tracial W^* -probability space. Let $(a_1, \dots, a_n)$ and $(b_1, \dots, b_m)$ be two n -tuples of self-adjoint elements in $\mathcal{N}$ such that, for all $c \in \mathcal{N}$ and $z \in \mathbb{C}^+$,*

$$\left\| \sum_{i=1}^n a_i c a_i - \sum_{i=1}^m b_i c b_i \right\|_{L^1} \leq K \|c\|_\infty.$$

Let $(N_1, \dots, N_{\max(n,m)})$ be a $\max(n, m)$ -tuple of independent Gaussian variables, and set

$$\mathbf{x} = \sum_{i=1}^n N_i a_i \quad \text{and} \quad \mathbf{y} = \sum_{i=1}^m N_i b_i.$$

Then, for any $z \in \mathbb{C}^+$,

$$\left| \mathbb{E}[\varphi(G_{\mathbf{x}}(z))] - \mathbb{E}[\varphi(G_{\mathbf{y}}(z))] \right| \leq \frac{K}{\text{Im}(z)^3}.$$

Proof. Without loss of generality, we replace $\mathbf{y}$ by $\sum_{i=1}^m \tilde{N}_i b_i$ where $(\tilde{N}_1, \dots, \tilde{N}_m)$ is m -tuple of independent Gaussian variables independent from $(N_1, \dots, N_n)$. We proceed by interpolation. Let us define $\mathbf{x}_t = \sqrt{t}\mathbf{x} + \sqrt{1-t}\mathbf{y}$ in such a way that $\mathbf{x}_1 = \mathbf{x}$ and $\mathbf{x}_0 = \mathbf{y}$. Then

$$\begin{aligned} \mathbb{E}[\varphi(G_{\mathbf{x}}(z))] - \mathbb{E}[\varphi(G_{\mathbf{y}}(z))] &= \mathbb{E} \int_0^1 \partial_t \varphi[G_{\mathbf{x}_t}(z)] \, dt \\ &= \mathbb{E} \int_0^1 \varphi \left[G_{\mathbf{x}_t}(z)^2 \left(\frac{\mathbf{x}}{2\sqrt{t}} - \frac{\mathbf{y}}{2\sqrt{1-t}} \right) \right] \, dt. \end{aligned}$$

We do an integration by part with respect to N_i and $\tilde{N}_i$ and we get

$$\begin{aligned} \mathbb{E}[\varphi(G_{\mathbf{x}}(z))] - \mathbb{E}[\varphi(G_{\mathbf{y}}(z))] &= \mathbb{E} \int_0^1 \varphi \left[G_{\mathbf{x}_t}(z)^2 \left(\sum_{i=1}^n a_i G_{\mathbf{x}_t}(z) a_i - \sum_{i=1}^m b_i G_{\mathbf{x}_t}(z) b_i \right) \right] \, dt \\ &\leq \frac{K}{\text{Im}(z)^3}. \end{aligned}$$

□

Proof of Theorem 4.6. The proof is based on the Gaussian interpolation technique in Lemma 4.7 that allows the change of covariance. To apply the lemma, we note first that

$$\frac{1}{n} \sum_{\substack{l \leq k \\ i, j}} \bar{x}_{kl} \otimes K_{ij} \cdot b \cdot \bar{x}_{kl} \otimes K_{ij} = \frac{1}{n} \sum_{l \leq k} \bar{x}_{kl} \, \text{tr}_N(b) \, \bar{x}_{kl}$$

and

$$\frac{1}{n} \sum_{\substack{l \leq k \\ j \leq i}} \bar{x}_{kl} \otimes^* a_{ij} \cdot b \cdot \bar{x}_{kl} \otimes^* a_{ij} = \frac{1}{n} \sum_{l \leq k} \bar{x}_{kl} \operatorname{tr}_N(b) \bar{x}_{kl} + \frac{1}{nN} \sum_{l \leq k} \bar{x}_{kl} \Theta(b) \bar{x}_{kl} - \frac{1}{nN} \sum_{\substack{l \leq k \\ i}} \bar{x}_{kl} \otimes E_{ii} \cdot b \cdot \bar{x}_{kl} \otimes E_{ii}$$

where $\Theta : M_N(\mathcal{M}) \rightarrow M_N(\mathcal{M})$ is the partial transpose $\Theta(M \otimes x) = M^t \otimes x$. We conclude by taking the L^1 -norm of their difference and by noting that

$$\left\| \frac{1}{nN} \sum_{l \leq k} \bar{x}_{kl} \Theta(b) \bar{x}_{kl} \right\|_{L^1} \leq \frac{4\|x_{11}\|_\infty^2}{N} \|\Theta(b)\|_{L^2} = \frac{4\|x_{11}\|_\infty^2}{N} \|b\|_{L^2} \leq \frac{4\|x_{11}\|_\infty^2}{N} \|b\|_\infty$$

and

$$\left\| \frac{1}{nN} \sum_{\substack{l \leq k \\ i}} \bar{x}_{kl} \otimes E_{ii} \cdot b \cdot \bar{x}_{kl} \otimes E_{ii} \right\|_{L^1} \leq \frac{4\|x_{11}\|_\infty^2}{N} \sum_i \|E_{ii}\|_{L^1} \|b\|_\infty \|E_{ii}\|_\infty = \frac{4\|x_{11}\|_\infty^2}{N} \|b\|_\infty.$$

□

We have reached now the last step in our proof in which we approximate the Gaussian operator-valued matrix Y by an operator-valued semicircular element. We note that the estimates given before hold for any $z \in \mathbb{C}^+$ and it is only in this very last step that we restrict it to a subset of the upper complex half-plane. This will become clearer in the proof of the following theorem.

Theorem 4.8. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space with $\mathcal{M}$ being a direct integral of finite dimensional factors of dimension $\leq d$. Let $\mathbf{x} = (x_{ij})_{1 \leq j \leq i \leq N}$ be a family of self-adjoint variables in $\mathcal{M}^n$, and let us denote by μ their mean. Consider a family $(N_{ij,kl})_{1 \leq j \leq i \leq N, 1 \leq l \leq k \leq N}$ of n^2 independent Gaussian random variables. We consider the random matrix Y of the previous theorem and the variance function $\eta(b) = \frac{1}{n} \sum_{j \leq i} \bar{x}_{ij} b \bar{x}_{ij}$. Let $\mathcal{B}$ be the smallest W^* -algebra which is closed under η and consider a centered semicircular variable S over $\mathcal{B}$ of variance $E_{\mathcal{B}}(SbS) = \eta(b)$. Letting A be the linear map defined in (4.1) then for any z such that $\operatorname{Im}(z) > \|\eta\|^{1/2}$, we have*

$$\|\mathbb{E}[\operatorname{tr}_N(G_Y(z))] - E_{\mathcal{B}}[G_S(z)]\|_{L^2(\mathcal{M}, \tau)} \leq \frac{2\pi d}{\operatorname{Im}(z)^2 - \|\eta\|} \left(\frac{\|x_{11}\|_\infty^3}{\operatorname{Im}(z)^2} \frac{1}{N} \right).$$

Remark 4.9. Exchangeability doesn't play any role in the above theorem which is valid for any family $\mathbf{x}$ in $\mathcal{M}^n$. The independent Gaussian random variables and the structure of the Wigner matrix are what allow such an approximation.

Proof. Let us denote by s the map $z \mapsto E_{\mathcal{B}}(G_S(z))$ and by g the map $z \mapsto \mathbb{E}[\operatorname{tr}_N(G_Y(z))]$. Remark that

$$\eta(s(z))s(z) - zs(z) + 1 = 0,$$

or in other words,

$$s(z) = G_{\eta(s(z))}(z).$$

We first note that for any z such that $\operatorname{Im}(z) > \|\eta\|^{1/2}$, if

$$g(z) = G_{\eta(g(z))}(z) + \delta,$$

then

$$\|\mathbb{E}[\operatorname{tr}_N(G_Y(z))] - E_{\mathcal{B}}(G_S(z))\|_{L^2(\mathcal{M}, \tau)} \leq (1 - \|\eta\| \operatorname{Im}(z)^{-2})^{-1} \|\delta\|_{L^2(\mathcal{M}, \tau)}.$$

Indeed, we have

$$\begin{aligned}
 \|g(z) - s(z)\|_{L^2(\mathcal{M}, \tau)} &= \|G_{\eta(g(z))}(z) - G_{\eta(s(z))}(z) + \delta\|_{L^2(\mathcal{M}, \tau)} \\
 &\leq \|G_{\eta(g(z))}(z)(\eta(g(z)) - \eta(s(z)))G_{\eta(s(z))}(z)\|_{L^2(\mathcal{M}, \tau)} + \|\delta\|_{L^2(\mathcal{M}, \tau)} \\
 &\leq \frac{1}{Im(z)^2} \|\eta\| \|g(z) - s(z)\|_{L^2(\mathcal{M}, \tau)} + \|\delta\|_{L^2(\mathcal{M}, \tau)} \\
 &\leq \frac{1}{Im(z)^2} \|\eta\| \|g(z) - s(z)\|_{L^2(\mathcal{M}, \tau)} + \|\delta\|_{L^2(\mathcal{M}, \tau)}.
 \end{aligned}$$

Thus in order to conclude the theorem, we have to show that

$$g(z) = G_{\eta(g(z))}(z) + \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right).$$

With this aim, we start by noticing that

$$1 = (z - Y)G_Y(z) = zG_Y(z) - YG_Y(z).$$

Note that

$$\mathbb{E}[YG_Y(z)] = \frac{1}{\sqrt{n}} \sum_{\substack{l \leq k \\ i, j}} \bar{x}_{kl} \mathbb{E}[N_{ij, kl} K_{ij} G_Y(z)]$$

and then integration by parts yields,

$$\begin{aligned}
 \mathbb{E}[YG_Y(z)] &= \frac{1}{\sqrt{n}} \sum_{\substack{l \leq k \\ i, j}} \bar{x}_{kl} \mathbb{E}[K_{ij} G_Y(z) K_{ij} \bar{x}_{kl} G_Y(z)] \\
 &= \frac{1}{\sqrt{n}} \sum_{l \leq k} \bar{x}_{kl} \mathbb{E}[\text{tr}_N(G_Y(z)) \bar{x}_{kl} G_Y(z)].
 \end{aligned}$$

Finally, $1 = zG_Y(z) - YG_Y(z)$ becomes

$$1 = zg(z) - \mathbb{E}[\eta(\text{tr}_N(G_Y(z))) \text{tr}_N(G_Y(z))].$$

Now, we shall show that we can replace $\mathbb{E}[\eta(\text{tr}_N(G_Y(z))) \text{tr}_N(G_Y(z))]$ by $\eta(g(z))g(z)$ with an error of order $\mathcal{O}_z(\frac{1}{\sqrt{N}})$. We proceed by multiplying the last identity by $G_{\eta(g(z))}(z)$ and get

$$\begin{aligned}
 G_{\eta(g(z))}(z) &= zG_{\eta(g(z))}(z)g(z) - G_{\eta(g(z))}(z)\mathbb{E}[\eta(\text{tr}_N(G_Y(z))) \text{tr}_N(G_Y(z))] \\
 &= (1 + G_{\eta(g(z))}(z)\eta(g(z)))g(z) - G_{\eta(g(z))}(z)\mathbb{E}[\eta(\text{tr}_N(G_Y(z))) \text{tr}_N(G_Y(z))]
 \end{aligned}$$

and from which we deduce that

$$\begin{aligned}
 G_{\eta(g(z))}(z) - g(z) &= G_{\eta(g(z))}(z)(\eta(g(z))g(z) - \mathbb{E}[\eta(\text{tr}_N(G_Y(z))) \text{tr}_N(G_Y(z))]) \\
 &= -G_{\eta(g(z))}(z)\mathbb{E}[\eta(\text{tr}_N(G_Y(z)))\Delta]
 \end{aligned}$$

where we denote by Δ the centered variable $\text{tr}_N(G_Y(z)) - \mathbb{E}[\text{tr}_N(G_Y(z))]$. Finally,

$$\begin{aligned}
 \|G_{\eta(g(z))}(z) - g(z)\|_{L^2} &\leq \mathbb{E} \|G_{\eta(g(z))}(z)\eta(\text{tr}_N(G_Y(z)))\Delta\|_{L^2} \\
 &\leq \frac{4\|x_{11}\|_{\infty}^2}{Im(z)^2} \mathbb{E}\|\Delta\|_{L^2} \\
 &\leq \frac{4\|x_{11}\|_{\infty}^2}{Im(z)^2} (\mathbb{E}\|\Delta\|_{L^2}^2)^{1/2}.
 \end{aligned}$$

We end the proof by proving that for any $z \in \mathbb{C}^+$,

$$(4.2) \quad \mathbb{E} \|\Delta\|_{L^2}^2 = \mathbb{E} \|\mathrm{tr}_N(G_Y(z)) - \mathbb{E}[\mathrm{tr}_N(G_Y(z))]\|_{L^2}^2 \leq \frac{\pi^2 d^2}{N^2} \frac{\|x_{11}\|_\infty^2}{\mathrm{Im}(z)^4}.$$

To prove (4.2) we shall apply a Gaussian concentration result of Pisier, which we state here:

Theorem 4.10 (Theorem 2.2, [26]). *Let F be a normed vector space and $\Phi : F \rightarrow \mathbb{R}$ be a convex function. Letting $\mathbf{N} = (N_1, \dots, N_n)$ be a standard Gaussian random vector in $\mathbb{R}^n$ then for any smooth and sufficiently integrable function $f : \mathbb{R}^n \rightarrow F$ such that $\mathbb{E}[f(\mathbf{N})] = 0$, we have*

$$\mathbb{E}[\Phi(f(\mathbf{N}))] \leq \mathbb{E}\left[\Phi\left(\frac{\pi}{2}\mathbf{N}' \cdot \nabla f(\mathbf{N})\right)\right],$$

where $\mathbf{N}'$ is an independent copy of $\mathbf{N}$.

Let $B : \mathbb{R}^{N^2n} \rightarrow M_N(\mathcal{M})$ be the map defined by

$$B((w_{ij,kl})_{1 \leq i,j \leq N, 1 \leq \ell \leq k \leq N}) = \frac{1}{\sqrt{n}} \sum_{i,j=1}^N \sum_{1 \leq \ell \leq k \leq N} w_{ij,kl} \bar{x}_{kl} \otimes K_{ij}.$$

Letting $\mathbf{N} = (N_{ij,kl})_{j \leq i, \ell \leq k}$, we note that $B(\mathbf{N}) = Y$. Then applying Theorem 4.10 for $f = \mathrm{tr}_N(G_{B(\cdot)}(z)) - \mathbb{E}[\mathrm{tr}_N(G_{B(\cdot)}(z))]$ and $\Phi = \|\cdot\|_{L^2(\mathcal{M}, \tau)}^2$ gives

$$\mathbb{E} \|\mathrm{tr}_N(G_Y(z)) - \mathbb{E}[\mathrm{tr}_N(G_Y(z))]\|_{L^2}^2 \leq \frac{\pi^2}{4} \mathbb{E} \|\mathrm{tr}_N(G_Y(z)Y'G_Y(z))\|_{L^2}^2$$

where $Y' = B(\mathbf{N}')$ with $\mathbf{N}'$ being an independent copy of $\mathbf{N}$. Now to control the right-hand term, we proceed by conditioning on $\mathbf{y}$:

$$\begin{aligned} & \mathbb{E} [\|\mathrm{tr}_N(G_Y(z)Y'G_Y(z))\|_{L^2}^2 | Y] \\ &= \tau(\mathbb{E}[\mathrm{tr}_N(G_Y(z)Y'G_Y(z)) \mathrm{tr}_N(G_Y(z)^*Y'G_Y(z)^*) | Y]) \\ &= \frac{1}{Nn} \sum_{\substack{l \leq k \\ i,j}} \tau(\mathrm{tr}_N(G_Y(z)(\bar{x}_{kl} \otimes E_{ij})G_Y(z)) \mathrm{tr}_N(G_Y(z)^*(\bar{x}_{kl} \otimes E_{ji})G_Y(z)^*)) \\ &= \frac{1}{N^3n} \sum_{\substack{l \leq k \\ i,j,m,n}} \tau(G_Y(z)_{mi}\bar{x}_{kl}G_Y(z)_{jm}(G_Y(z)^*)_{nj}\bar{x}_{kl}(G_Y(z)^*)_{in}) \\ &= \frac{1}{N^2n} \sum_{l \leq k} \tau(\mathrm{tr}_N(\Theta(G_Y(z))(I_N \otimes \bar{x}_{kl})\Theta(G_Y(z))\Theta(G_Y(z)^*)(I_N \otimes \bar{x}_{kl})\Theta(G_Y(z)^*))) \end{aligned}$$

where $\Theta : M_N(\mathcal{M}) \rightarrow M_N(\mathcal{M})$ is the partial transpose $\Theta(M \otimes x) = M^t \otimes x$. As a consequence, using Lemma 4.11,

$$\mathbb{E} [\|\mathrm{tr}_N(G_Y(z)Y'G_Y(z))\|_{L^2}^2 | Y] \leq \frac{4d^2}{N^2} \frac{\|x_{11}\|_\infty^2}{\mathrm{Im}(z)^4}.$$

□

Lemma 4.11. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space with $\mathcal{M}$ being a direct integral of finite dimensional factors of dimension $\leq d$. Then the partial transpose $\Theta : M_N(\mathcal{M}) \rightarrow M_N(\mathcal{M})$ is bounded. More precisely, we have*

$$\|\Theta(M)\|_2 = \|M\|_2$$

and

$$\|\Theta(M)\|_\infty \leq d \cdot \|M\|_\infty.$$

Proof. It is a consequence of [36]. $\square$

4.3. Infinite exchangeable entries. It is well-known by an extended de Finetti theorem [13], that an *infinite* family of identically distributed and conditionally independent elements over a tail algebra is exchangeable. In the setting of conditionally free entries, the asymptotic freeness of Wigner matrices and their convergence to operator-valued semicircular elements were studied by Liu [18]. This section is in this direction where we consider an infinite family of exchangeable elements and prove the convergence to an operator-valued semicircular over the tail algebra. We provide moreover explicit rates of convergence for the associated Cauchy transforms.

Proposition 4.12. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability spaces and let $\mathbf{x} = (x_{ij})_{1 \leq j \leq i < +\infty}$ be an infinite family of exchangeable variables in $\mathcal{M}$. Let $\mathcal{M}_{\text{tail}}$ be the tail subalgebra and $E : \mathcal{M} \rightarrow \mathcal{M}_{\text{tail}}$ be its conditional expectation. Consider a centered semicircular variable S over $\mathcal{M}_{\text{tail}}$ of variance $E(SbS) = E((x_{11} - E(x_{11}))b(x_{11} - E(x_{11})))$. Fix $N \in \mathbb{N}$ and let A be the linear map defined in (4.1). Then for any z such that $\text{Im}(z) > \|x_{11}\|_\infty$, we have*

$$(\tau \otimes \text{tr}_N)[G_{A(\mathbf{x})}(z)] - \tau[G_S(z)] = O_z(1/\sqrt{N}).$$

Before proceeding to the proof, we give the following lemma which allows passing to a smaller subalgebra, say $\mathcal{B}$, and exhibiting a $\mathcal{B}$ -valued semicircular element in the limit only under some factorization conditions on the conditional expectation over $\mathcal{B}$.

Lemma 4.13. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space and let $\mathbf{x} = (x_i)_{1 \leq i \leq n}$ be a family of exchangeable self-adjoint variables in $\mathcal{M}$. Set $\bar{x}_i = x_i - \mu$ where $\mu = \frac{1}{n} \sum_{1 \leq i \leq n} x_i$ and consider their variance function*

$$\eta_n : \mathcal{M} \rightarrow \mathcal{M}, \quad \eta_n(b) = \frac{1}{n} \sum_{i=1}^n \bar{x}_i b \bar{x}_i.$$

Let $\mathcal{B}_n$ be the smallest W^ -algebra which is closed under η_n and consider a centered semicircular variable S_n over $\mathcal{B}_n$ of variance $E_{\mathcal{B}_n}(S_n b S_n) = \eta_n(b)$. We consider a von Neumann subalgebra $\mathcal{B}$ and $E : \mathcal{M} \rightarrow \mathcal{B}$ its conditional expectation. Assume that, whenever i, j are distinct, for all $b, c \in \mathcal{B}$,*

$$E(x_i b x_j) = E(x_1) b E(x_1) \quad \text{and} \quad E(x_i b x_i x_k c x_k) = E(x_1 b x_1) E(x_1 c x_1).$$

Consider a centered semicircular variable S over $\mathcal{B}$ of variance

$$E(SbS) = E((x_1 - E(x_1))b(x_1 - E(x_1))) := \eta(b).$$

Then, for any z such that $\text{Im}(z) > \|x_1\|_\infty$, we have

$$\tau[G_{S_n}(z)] - \tau[G_S(z)] = O_z(1/\sqrt{n}).$$

Proof. We have

$$G_S(z) = G_{\eta(G_S(z))}(z)$$

and

$$G_{S_n}(z) = G_{\eta_n(G_{S_n}(z))}(z) = G_{\eta_n(G_S(z))}(z) + G_{\eta_n(G_{S_n}(z))}(z) \cdot \eta_n(G_{S_n}(z) - G_S(z)) \cdot G_{\eta_n(G_{S_n}(z))}(z).$$

Noting that $\eta(b) = E(\eta_n(b))$, we get

$$\|G_{S_n}(z) - G_S(z)\|_{L^2} \leq \frac{1}{Im(z)^2} \|\eta_n\| \|G_{S_n}(z) - G_S(z)\|_{L^2} + \|G_{\eta_n(G_S(z))} - G_{E[\eta_n(G_S(z))]}(z)\|_{L^2}.$$

Thus, for $Im(z) > \|\eta_n\|^{1/2}$, we have

$$\begin{aligned} \|G_{S_n}(z) - G_S(z)\|_{L^2} &\leq \frac{1}{1 - \|\eta_n\| Im(z)^{-2}} \|G_{\eta_n(G_S(z))} - G_{E[\eta_n(G_S(z))]}(z)\|_{L^2} \\ &\leq \frac{1}{Im(z)^2 - \|\eta_n\|} \|\eta_n(G_S(z)) - E[\eta_n(G_S(z))]\|_{L^2}. \end{aligned}$$

But for all $b \in \mathcal{B}$, we have

$$\begin{aligned} \|\eta_n(b) - E[\eta_n(b)]\|_{L^2} &= \left\| \frac{1}{n} \sum_{i=1}^n \bar{x}_i b \bar{x}_i - E[(x_1 - E(x_1))b(x_1 - E(x_1))] \right\|_{L^2} \\ &\leq \left\| \frac{1}{n} \sum_{i=1}^n x_i b x_i - E[x_1 b x_1] \right\|_{L^2} + \|\mu b \mu - E[x_1] b E[x_1]\|_{L^2}. \end{aligned}$$

We start by controlling the first term in the last inequality and obtain

$$\begin{aligned} \left\| \frac{1}{n} \sum_{i=1}^n x_i b x_i - E[x_1 b x_1] \right\|_{L^2}^2 &= \tau \left(\frac{1}{n^2} \sum_{i,j=1}^n x_i b x_i x_j b x_j - E[x_1 b x_1] \right) \\ &= \tau \left(\left(\frac{n(n-1)}{n^2} - 1 \right) (E[x_1 b x_1])^2 + \frac{1}{n^2} \sum_{i=1}^n x_i b x_i^2 b x_i \right) \\ &\leq \frac{2}{n} \|x_1\|_\infty^4 \|b\|_\infty^2 = \mathcal{O}\left(\frac{1}{n}\right) \end{aligned}$$

where in the first equality we have used the property that $E(x_i b x_i x_j b x_j) = E(x_1 b x_1) E(x_1 b x_1)$ for $i \neq j$ and the fact that $\tau(aE[a]) = \tau \circ E[aE[a]] = \tau(E[a])^2$. Therefore we get

$$\left\| \frac{1}{n} \sum_{i=1}^n x_i b x_i - E[x_1 b x_1] \right\|_{L^2} = \mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$$

To control the second term we write

$$\begin{aligned} \|\mu b \mu - E[x_1] b E[x_1]\|_{L^2} &= \|(\mu - E[x_1]) b \mu + E[x_1] b (\mu - E[x_1])\|_{L^2} \\ &\leq 2\|x_1\|_\infty \|b\|_\infty \|\mu - E[x_1]\|_{L^2} = \mathcal{O}\left(\frac{1}{\sqrt{n}}\right) \end{aligned}$$

where the estimate on $\|\mu - E[x_1]\|_{L^2}$ is obtained in a similar way by using the property that $E(x_i b x_j) = E(x_1) b E(x_1)$ whenever $i \neq j$. $\square$

Proof of Proposition 4.12. Theorem 4.3 ensures that

$$(\tau \otimes \text{tr}_N)[G_{A(\mathbf{x})}(z)] - \tau[G_{S_N}(z)] = O_z(1/\sqrt{N}),$$

where S_N is a centered semicircular variable of variance $E_B(S_N b S_N) = \eta(b) = \frac{1}{n} \sum_{j \leq i} \bar{x}_{ij} b \bar{x}_{ij}$ where $\bar{x}_{ij} = x_{ij} - \frac{1}{n} \sum_{1 \leq \ell \leq k \leq N} X_{k\ell}$ and $n = \frac{N}{2}(N+1)$. By using Lemma 4.13, we prove that

$$|\tau[G_{S_N}(z)] - \tau[G_S(z)]| = \mathcal{O}_z\left(\frac{1}{N}\right).$$

Indeed, by the conditional independence of the x_{ij} 's with respect to E [13], we have for any $b, c \in \mathcal{M}_{tail}$

$$E(x_{ij}bx_{kl}) = E(x_{11})bE(x_{11}) \quad \text{and} \quad E(x_{ij}bx_{ij}x_{kl}cx_{kl}) = E(x_{11}bx_{11})E(x_{11}cx_{11})$$

whenever $x_{ij} \neq x_{kl}$. $\square$

5. OPERATOR-VALUED WISHART MATRICES

Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space and let $\mathbf{h} = (h_{ij})_{1 \leq i, j \leq N}$ be a family of elements in $\mathcal{M}$. Consider the $N \times N$ matrix H in $(M_N(\mathcal{M}), \tau \otimes \text{tr}_N)$ given by

$$H := H_N = \begin{pmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & & \vdots \\ h_{N1} & \cdots & h_{NN} \end{pmatrix} \in M_N(\mathcal{M}).$$

We are interested in studying the $N \times N$ operator-valued Wishart matrix $\frac{1}{N}HH^*$. This is an interesting model in random matrix theory and is very well understood in the classical setting of commutative entries. However, not much is known in the operator-valued case despite its significance importance for many applications. We mention, among others, wireless communication systems whose channels have such a Hermitian block structure [29]. Although our result is of different nature, we mention the work of Bryc [5] in which the author considers a family of q -Wishart orthogonal matrices.

In this section, we consider operator-valued Wishart matrices with free or exchangeable entries and describe their distribution by giving quantitative estimates on the associated Cauchy transforms. We will use the same approach as before to study this model. However, the map $B : (h_{ij})_{1 \leq i, j \leq N} \mapsto B((h_{ij})_{1 \leq i, j \leq N}) = \frac{1}{N}HH^*$ is not linear and thus our noncommutative Lindeberg method in Section 3 is not directly applicable. To fix this up, we shall rather consider the Hermitian matrix

$$X = \frac{1}{\sqrt{N}} \begin{pmatrix} 0 & H \\ H^* & 0 \end{pmatrix} \in M_{2N}(\mathcal{A})$$

and note that a direct application of the Schur complement formula yields for any $z \in \mathbb{C}^+$

$$(5.1) \quad (\tau_{\mathcal{A}} \otimes \text{tr}_{2N})[G_X(z)] = 2z \cdot (\tau_{\mathcal{A}} \otimes \text{tr}_N)[G_{\frac{1}{N}HH^*}(z^2)].$$

This reduces the study to the matrix X which is in turn equivalent, through a permutation of the entries, to the Hermitian matrix $\mathbf{X} = (\frac{1}{\sqrt{N}}x_{ij})_{1 \leq i, j \leq N}$ with

$$(5.2) \quad x_{ij} = \begin{pmatrix} 0 & h_{ij} \\ h_{ji}^* & 0 \end{pmatrix} \in M_2(\mathcal{M}).$$

Note that $\mathbf{X}$ is an operator-valued Wigner type matrix with entries in $M_2(\mathcal{M})$. This allows us to apply the results in Section 4 on Wigner matrices and extend them to Wishart matrices.

5.1. Free entries. We consider first matrices with free centered entries that don't need to be identically distributed.

Theorem 5.1. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space and $\mathbf{h} = (h_{ij})_{1 \leq i, j \leq N}$ be a family of free centered elements in $\mathcal{M}$ such that $\varphi(h_{ij}^*h_{ij}) = \varphi(h_{ji}^*h_{ji})$. Let $\mathbf{c} = (c_{ij})_{1 \leq i, j \leq N}$ be*

a family of free centered circular elements in $\mathcal{M}$ such that $\varphi(c_{ij}^*c_{ij}) = \varphi(h_{ij}^*h_{ij})$ for every $1 \leq i, j \leq N$. Then for any $z \in \mathbb{C}^+$,

$$|(\tau \otimes \text{tr}_N)[G_{\frac{1}{N}HH^*}(z)] - (\tau \otimes \text{tr}_N)[G_{\frac{1}{N}CC^*}(z)]| = \mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$$

where C is the $N \times N$ operator valued matrix given by $C = [(c_{ij})_{1 \leq i, j \leq N}]$.

Proof. By (5.1) it is equivalent to prove for any $z \in \mathbb{C}^+$

$$|(\tau \otimes \text{tr}_{2N})[G_X(z)] - (\tau \otimes \text{tr}_{2N})[G_Y(z)]| = \mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$$

where X and Y are the Hermitian block matrices given by

$$X = \frac{1}{\sqrt{N}} \begin{pmatrix} 0 & H \\ H^* & 0 \end{pmatrix} \quad \text{and} \quad Y = \frac{1}{\sqrt{N}} \begin{pmatrix} 0 & C \\ C^* & 0 \end{pmatrix}.$$

Now X and Y are in turn equivalent, by a permutation of the entries, to $\mathbf{X} = (\frac{1}{\sqrt{N}}x_{ij})_{1 \leq i, j \leq N}$ and $\mathbf{Y} = (\frac{1}{\sqrt{N}}y_{ij})_{1 \leq i, j \leq N}$ with

$$x_{ij} = \begin{pmatrix} 0 & h_{ij} \\ h_{ji}^* & 0 \end{pmatrix} \quad \text{and} \quad y_{ij} = \begin{pmatrix} 0 & c_{ij} \\ c_{ji}^* & 0 \end{pmatrix} \in M_2(\mathcal{M}).$$

Setting $(\mathcal{A}, \varphi) := (M_2(\mathcal{M}), \tau \otimes \text{tr}_2)$ and recalling the linear map A defined in (4.1), we note that $\mathbf{X} = A((x_{ij})_{1 \leq j \leq i \leq N})$ and $\mathbf{Y} = A((y_{ij})_{1 \leq j \leq i \leq N})$. We only need to check now that Theorem 4.1 holds in this case and then the remaining part of the proof would be a direct application of it. We start by noting that for any i and $j < i$, y_{ii} and y_{ij} are respectively semicircular and circular elements in $\mathcal{A}$. Indeed, for any $k \in \mathbb{N}$

$$\varphi(y_{ii}^{2k}) = \tau(s_{ii}^{2k}) \quad \text{and} \quad \varphi((y_{ij}^*y_{ij})^{2k}) = \tau(s_{ij}^{2k})$$

where s_{ii} and s_{ij} are semicircular elements in $\mathcal{M}$ of variance $\tau(c_{ii}^*c_{ii})$ and $\tau(c_{ij}^*c_{ij})$ respectively. The remaining conditions follow mainly by freeness and the fact that the variables are centered and have same variance. $\square$

Remark 5.2. If $\tau(c_{ij}c_{ij}^*) = 1$ for all $i, j \in \{1, \dots, N\}$ then $\frac{1}{\sqrt{N}}C$ is itself a circular element in $\mathcal{M}$ and hence $\frac{1}{N}CC^*$ is a free Poisson element in $\mathcal{M}$. This can be proved using similar ideas as in [20, Chapter 6, Theorem 11].

Exchangeable entries. We study in this section the behavior of operator-valued Wigner-type matrices with exchangeable entries. We state now the main result:

Theorem 5.3. *Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space with $\mathcal{M}$ being a direct integral of finite dimensional factors of dimension $\leq d$. Let $\mathbf{h} = (h_{ij})_{1 \leq i, j \leq N}$ be a family of exchangeable variables in $\mathcal{M}^n$ with $n = N(N+1)/2$. Let $\mathbf{x} = (x_{ij})_{1 \leq i, j \leq N}$ with x_{ij} as defined as in (5.2), set $\bar{x}_{ij} = x_{ij} - \mu$ where $\mu = \frac{1}{n} \sum_{1 \leq j \leq i \leq N} x_{ij}$ and consider their variance function*

$$\eta : M_2(\mathcal{M}) \rightarrow M_2(\mathcal{M}), \quad \eta(b) = \frac{1}{n} \sum_{j \leq i} \bar{x}_{ij} b \bar{x}_{ij}.$$

Let $\mathcal{B}$ be the smallest W^* -algebra which is closed under η and consider a centered $\mathcal{B}$ -valued semicircular variable S of variance $E_{\mathcal{B}}(SbS) = \eta(b)$. Then for any z such that $\text{Im}(z^{1/2}) > \|h_{11}\|_{\infty}$, we have

$$|(\tau \otimes \text{tr}_N)[G_{\frac{1}{N}HH^*}(z)] - \tau[G_S(z)]| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right).$$

Proof. As explained above in this Section, we shall pass the study to the Wigner-type matrix $\mathbf{X} = (\frac{1}{\sqrt{N}}x_{ij})_{1 \leq i, j \leq N}$ and then apply results of Section 4.2. However, in this case, the diagonal and off-diagonal entries of $\mathbf{X}$ are not exchangeable in $(M_2(\mathcal{M}), \tau \otimes \text{tr}_2)$. For this, we shall replace the diagonal entries by some elements in $M_2(\mathcal{M})$ that are exchangeable with the off-diagonal entries so that this fits in the setting of Theorem 4.3 and prove that this replacement will not affect the asymptotic behavior.

Let $(\tilde{x}_{ii})_{1 \leq i \leq N}$ be a family of exchangeable elements that are also exchangeable with $(x_{ij})_{1 \leq j < i \leq N}$. Let $\tilde{\mathbf{x}}$ be the family defined as $\mathbf{x}$ but with the diagonal entries replaced by the $\tilde{x}_{ii}$'s. The resolvent identity gives

$$G_{A(\mathbf{x})}(z) - G_{A(\tilde{\mathbf{x}})}(z) = G_{A(\mathbf{x})}(z) \left(\frac{1}{2\sqrt{N}} \sum_{i=1}^N (x_{ii} - \tilde{x}_{ii}) \otimes^* E_{ii} \right) G_{A(\tilde{\mathbf{x}})}(z)$$

and hence

$$\begin{aligned} |(\tau \otimes \text{tr}_N)[G_{A(\mathbf{x})}(z)] - (\tau \otimes \text{tr}_N)[G_{A(\tilde{\mathbf{x}})}(z)]| &\leq \frac{1}{2\sqrt{N} \text{Im} z^2} \left\| \sum_{i=1}^N (x_{ii} - \tilde{x}_{ii}) \otimes^* E_{ii} \right\|_{L_1} \\ &\leq \frac{2}{\sqrt{N} \text{Im} z^2} \|h_{11}\|_{\infty}. \end{aligned}$$

Now let $\tilde{\mu} = \frac{1}{n} \sum_{1 \leq j < i \leq N} x_{ij} + \frac{1}{n} \sum_i \tilde{x}_{ii}$ and set $\hat{x}_{ii} = \tilde{x}_{ii} - \tilde{\mu}$ and $\hat{x}_{ij} = x_{ij} - \tilde{\mu}$ for $j < i$. Therefore applying Theorem 4.3 to $A(\tilde{\mathbf{x}})$ we get for any z such that $\text{Im}(z) > \|h_{11}\|_{\infty}$,

$$|(\tau \otimes \text{tr}_N)[G_{A(\tilde{\mathbf{x}})}(z)] - \tau[G_{\tilde{S}}(z)]| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right)$$

where $\tilde{S}$ is a centered $\mathcal{B}$ -valued semicircular variable of variance $E_{\mathcal{B}}(\tilde{S}b\tilde{S}) = \tilde{\eta}(b)$ with

$$\tilde{\eta} : M_2(\mathcal{M}) \rightarrow M_2(\mathcal{M}), \quad \tilde{\eta}(b) = \frac{1}{n} \sum_{j \leq i} \hat{x}_{ij} b \hat{x}_{ij}.$$

Now we have

$$G_S(z) = G_{\eta(G_S(z))}(z) \quad \text{and} \quad G_{\tilde{S}}(z) = G_{\tilde{\eta}(G_{\tilde{S}}(z))}(z)$$

and then following the same steps as in the proof of Lemma 4.13 we get

$$\|G_S(z) - G_{\tilde{S}}(z)\|_{L^2} \leq \frac{1}{\text{Im}(z)^2 - \|\eta\|} \|\eta(G_S(z)) - \tilde{\eta}(G_S(z))\|_{L^2}.$$

To end the proof, we still need check that for all $b \in \mathcal{B}$

$$\|\eta(b) - \tilde{\eta}(b)\|_{L^2} = O\left(\frac{1}{\sqrt{n}}\right) = O\left(\frac{1}{N}\right).$$

Indeed, let us note that

$$\|\mu - \tilde{\mu}\|_{L^2} \leq \frac{1}{n} \sum_{i=1}^N \|x_{ii} - \tilde{x}_{ii}\|_{L^2} \leq 2 \frac{N}{n} \|x_{ii}\|_{\infty} = O\left(\frac{1}{\sqrt{n}}\right) = O\left(\frac{1}{N}\right).$$

and thus we have

$$\|\mu b \mu - \tilde{\mu} b \tilde{\mu}\|_{L^2} = \|(\mu - \tilde{\mu})b\mu - \mu b(\mu - \tilde{\mu})\|_{L^2} \leq 2\|b\|_{\infty} \|x_{11}\|_{\infty} \|\mu - \tilde{\mu}\|_{L^2} = O\left(\frac{1}{\sqrt{n}}\right)$$

which then yields that

$$\begin{aligned} \|\eta(b) - \tilde{\eta}(b)\|_{L^2} &= \left\| \frac{1}{n} \sum_{j < i} \bar{x}_{ij} b \bar{x}_{ij} - \frac{1}{n} \sum_{j < i} \hat{x}_{ij} b \hat{x}_{ij} \right\|_{L^2} + O\left(\frac{1}{\sqrt{n}}\right) \\ &\leq 2\|b\|_{\infty} \|x_{11}\|_{\infty} \|\mu - \tilde{\mu}\|_{L^2} + \|\mu b \mu - \tilde{\mu} b \tilde{\mu}\|_{L^2} + O\left(\frac{1}{\sqrt{n}}\right) = O\left(\frac{1}{\sqrt{n}}\right). \end{aligned}$$

□

6. APPLICATIONS: SPECTRA OF LARGE BLOCK MATRICES

The aim of this section is to give some applications of our results for random block matrices. Indeed, many types of random matrices fit well in the setting of noncommutative *-probability spaces. As examples of applications, we consider matrices with independent blocks and matrices in which the blocks are themselves correlated but have i.i.d. entries. We give quantitative estimates on the associated Cauchy transforms which lead in some cases to convergence in Kolmogorov distance. We shall only give, using the results of Section 4, applications to matrices having some Wigner structure but these applications can also be extended to the Wishart case by applying the results of Section 5 instead.

6.1. Random matrices with i.i.d. blocks. Since i.i.d. matrices are exchangeable, random block matrices with i.i.d blocks fit instantly in our framework. We generalize the result in [10] by considering matrices that are not necessary bounded in norm and give quantitative estimates for the Cauchy transforms.

Theorem 6.1. *Let $\{A_{ij} = (a_{kl}^{(ij)})_{k,l=1}^d : 1 \leq j \leq i \leq N\}$ be a family of independent and identically distributed $d \times d$ random matrices that are such that $\mathbb{E}[|a_{kl}^{(ij)}|^3] < \infty$. Let X_N be the $Nd \times Nd$ block Wigner matrix given by $X_N = A((A_{ij})_{1 \leq j \leq i \leq N})$ where A is the mapping defined in (4.1). Then, as $N \rightarrow \infty$, the matrix X_N has a limiting eigenvalue distribution whose Cauchy transform g is determined by $g(z) = \text{tr}_d(G(z))$, where G is an $M_d(\mathbb{C})$ -valued analytic function on $\mathbb{C}^+$ uniquely determined by the requirement that, $zG(z) \rightarrow 1$ when $|z| \rightarrow \infty$, and that, for $z \in \mathbb{C}^+$,*

$$zG(z) = 1 + \eta(G(z)) \cdot G(z)$$

where $\eta : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is the covariance mapping

$$\eta(B) = \mathbb{E}[(A_{11} - \mathbb{E}[A_{11}])B(A_{11} - \mathbb{E}[A_{11}])].$$

Moreover, there exists $K > 0$ such that for any z with $\text{Im}(z) > K$, we have

$$|\mathbb{E}[\text{tr}_{dN}(G_{X_N}(z))] - g(z)| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right).$$

Proof. We remark again that the blocks A_{ij} are i.i.d. and thus exchangeable. However, our Theorem 4.3 is only applicable if the block matrices are bounded which is not necessarily the case here. In order to apply our results, we will approximate first X_N by a matrix $Y_N = A((B_{ij})_{1 \leq j \leq i \leq N})$ where the blocks $B_{ij} = (b_{kl}^{(ij)})_{k,l=1}^d : 1 \leq j \leq i \leq N$ are i.i.d. matrices that are bounded in norm by some $K > 0$ and have the same mean and covariance structure as the A_{ij} 's, i.e. $\text{Cov}(b_{pq}^{(ij)}, b_{rs}^{(ij)}) = \text{Cov}(a_{pq}^{(ij)}, a_{rs}^{(ij)})$ (this can be done for example by taking a linear map decorrelating the $a_{kl}^{(ij)}$'s and replacing the corresponding random variables by

independent two-point distributions). Note that X_N and Y_N have respectively the same distribution as

$$\sum_{1 \leq j \leq i \leq N} A_{ij} \otimes^* e_{ij} \quad \text{and} \quad \sum_{1 \leq j \leq i \leq N} B_{ij} \otimes^* e_{ij},$$

where $e_{ii} = \frac{1}{2\sqrt{N}}E_{ii}$ and $e_{ij} = \frac{1}{\sqrt{N}}E_{ij}$ for $j < i$ and $(E_{ij})_{1 \leq i, j \leq N}$ are the canonical $N \times N$ matrices. Let us now denote respectively by $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ the n -tuples $\{A_{ij} \otimes^* e_{ij} : 1 \leq j \leq i \leq N\}$ and $\{B_{ij} \otimes^* e_{ij} : 1 \leq j \leq i \leq N\}$ of self-adjoint random elements in $\mathcal{M}$. Setting for any $i \in \{1, \dots, n\}$,

$$\mathbf{z}_i = x_1 + \dots + x_i + y_{i+1} + \dots + y_n$$

and

$$\mathbf{z}_i^0 = x_1 + \dots + x_{i-1} + y_{i+1} + \dots + y_n$$

then X_N and Y_N have the same distributions as $\mathbf{z}_n$ and $\mathbf{z}_0$ respectively. Therefore, applying the noncommutative Lindeberg method in Theorem 3.1, we get for any $z \in \mathbb{C}^+$,

$$\left| \mathbb{E}[\text{tr}_{dN}(G_{X_N}(z))] - \mathbb{E}[\text{tr}_{dN}(G_{Y_N}(z))] \right| \leq \sum_{i=1}^n (|\mathbb{E}[P_i]| + |\mathbb{E}[Q_i]| + |\mathbb{E}[R_i]|).$$

where, for any $i \in \{1, \dots, n\}$,

$$P_i = \text{tr}_d \otimes \text{tr}_N [G_{\mathbf{z}_i^0}(z)(x_i - y_i)G_{\mathbf{z}_i^0}(z)],$$

$$Q_i = \text{tr}_d \otimes \text{tr}_N [G_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)x_iG_{\mathbf{z}_i^0}(z)] - \text{tr}_d \otimes \text{tr}_N [G_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)y_iG_{\mathbf{z}_i^0}(z)],$$

and

$$R_i = \text{tr}_d \otimes \text{tr}_N [G_{\mathbf{z}_i}(z)(x_iG_{\mathbf{z}_i^0}(z))^3] - \tau [G_{\mathbf{z}_{i-1}}(z)(y_iG_{\mathbf{z}_i^0}(z))^3].$$

As the B_{ij} 's are independent and as the first and second moments of their entries match with those of the A_{ij} 's, we get that $\mathbb{E}[P_i] = 0$ and $\mathbb{E}[Q_i] = 0$. Furthermore, we have for any $i \in \{1, \dots, n\}$

$$\begin{aligned} |\mathbb{E}[R_i]| &\leq \left| \mathbb{E} \text{tr}_d \otimes \text{tr}_N [G_{\mathbf{z}_i}(z)(x_iG_{\mathbf{z}_i^0}(z))^3] \right| + \left| \mathbb{E} \text{tr}_d \otimes \text{tr}_N [G_{\mathbf{z}_i}(z)(y_iG_{\mathbf{z}_i^0}(z))^3] \right| \\ &\leq \frac{1}{\text{Im}(z)^4} \mathbb{E} \text{tr}_d \otimes \text{tr}_N [|x_i|^3] + \frac{1}{\text{Im}(z)^4} \mathbb{E} \text{tr}_d \otimes \text{tr}_N [|y_i|^3] \\ &\leq \frac{1}{\text{Im}(z)^4} \frac{\mathbb{E} \text{tr}_d[|B_{11}|^3]}{N^{5/2}} + \frac{1}{\text{Im}(z)^4} \frac{K^3}{N^{5/2}}. \end{aligned}$$

Finally, summing over $i = 1, \dots, N(N+1)/2$, we get

$$\left| \mathbb{E}[\text{tr}_{dN}(G_{X_N}(z))] - \mathbb{E}[\text{tr}_{dN}(G_{Y_N}(z))] \right| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right).$$

The study is thus reduced to Y_N and Theorem 4.3 now directly ensures that for any $z \in \mathbb{C}^+$ such that $\text{Im}(z) > K$

$$\left| \mathbb{E} \text{tr}_d \otimes \text{tr}_N [G_{Y_N}(z)] - \tau[G_{S_N}(z)] \right| = \mathcal{O}_z(1/\sqrt{N}),$$

where S_N is a centered semicircular variable over the subalgebra of $d \times d$ random matrices of covariance $\eta_n(b) = \frac{1}{n} \sum_{j \leq i} \bar{B}_{ij} b \bar{B}_{ij}$ where $\bar{B}_{ij} = B_{ij} - \frac{1}{n} \sum_{\ell \leq k} B_{k\ell}$ and $n = N(N+1)/2$. It remains to prove that

$$\left| \tau[G_{S_N}(z)] - \tau[G(z)] \right| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right).$$

Indeed, this follows directly by Lemma 4.13 which even guarantees that the rate of convergence is of the order of N^{-1} . $\square$

6.2. Matrices with correlated blocks. We consider in this section, block random matrices in which the blocks are correlated. As a first example, we consider in Theorem 6.2 matrices with a Wigner block structure in which the blocks can be correlated but their entries are i.i.d. random variables and have finite third moments. We consider general correlation functions and give quantitative estimates for the Cauchy transform. In this way, we generalize the results in [29, 25] in which the limiting distribution was described without any rates of convergence. We then give in Theorem 6.3 a concrete example on circulant block matrices for which the distribution of the operator-valued semicircular can be explicitly computed and for which the rate of convergence of the Cauchy transform implies in turn convergence in Kolmogorov distance.

Theorem 6.2. *Fix $d \geq 1$ and consider the $Nd \times Nd$ block matrix X_N defined by*

$$X_N = \frac{1}{\sqrt{d}} \begin{pmatrix} A^{(11)} & \dots & A^{(1d)} \\ \vdots & \ddots & \vdots \\ A^{(d1)} & \dots & A^{(dd)} \end{pmatrix}$$

where, for any $1 \leq j \leq i \leq d$, $A^{(ji)} = (A^{(ij)})^*$ and $A^{(ij)} = (\frac{1}{\sqrt{N}} a_{rp}^{(ij)})_{r,p=1}^N$ are $N \times N$ random matrices having i.i.d entries and such that for any $i, j, k, \ell = 1, \dots, d$

$$\text{Cov}(a_{11}^{(ij)}, a_{11}^{(kl)}) = \sigma(i, j; k, l) \quad \text{and} \quad \mathbb{E}[|a_{11}^{(ij)}|^3] < \infty.$$

Then, for $N \rightarrow \infty$, the matrix X_N has a limiting eigenvalue distribution whose Cauchy transform g is determined by $g(z) = \text{tr}_d(G(z))$, where G is an $M_d(\mathbb{C})$ -valued analytic function on $\mathbb{C}^+$, which is uniquely determined by the requirement that, $zG(z) \rightarrow 1$ when $|z| \rightarrow \infty$, and that, for $z \in \mathbb{C}^+$,

$$zG(z) = 1 + \eta(G(z)) \cdot G(z)$$

where $\eta : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is the covariance mapping

$$\eta(B)_{ij} = \sum_{k,l=1}^d \frac{1}{d} \sigma(i, k; l, j) B_{kl}.$$

Moreover, there exists $K > 0$ such that for any z with $\text{Im}(z) > K$, we have

$$|\mathbb{E}[\text{tr}_{dN}(G_{X_N}(z))] - g(z)| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right).$$

Note that when the cumulative distribution of the operator-valued semicircular (whose Cauchy transform is g) is Lipschitz, the estimate on the Cauchy transform gives quantitative estimates on the Kolmogorov distance [32, Theorem A.2].

Proof. Let us write

$$X_N = \frac{1}{\sqrt{d}} \sum_{i,j} A^{(ij)} \otimes E_{ij} = \frac{1}{\sqrt{dN}} \sum_{\substack{i,j \\ r,p}} a_{rp}^{(ij)} \cdot E_{rp} \otimes E_{ij} = \frac{1}{\sqrt{N}} \sum_{r,p} E_{rp} \otimes A_{rp}$$

where $A_{rp} = A_{pr}^*$ and the blocks $A_{rp} = \frac{1}{\sqrt{d}}(a_{rp}^{(ij)})_{i,j=1}^d$ are $d \times d$ random matrices with covariance $\sigma(i, j; k, l)/d$. As a consequence, X_N has the same distribution as the matrix

$$\frac{1}{\sqrt{N}} \begin{pmatrix} A_{11} & \cdots & A_{1N} \\ \vdots & \ddots & \vdots \\ A_{N1} & \cdots & A_{NN} \end{pmatrix}.$$

We note that the off-diagonal blocks are identically distributed and thus exchangeable; however, they are not exchangeable with the diagonal blocks. So in order to apply our previous results, we shall replace first the diagonal blocks by blocks having the same distribution as the off-diagonal ones. This can be easily done as in the proof of Theorem 5.3 and will thus be omitted. Therefore, we can assume now that the blocks A_{rp} are all exchangeable for $1 \leq p \leq r \leq N$ and the rest of the proof follows as in Theorem 6.1. $\square$

We give another possible application of our results to self-adjoint block circulant matrices. Let $(\mathcal{M}, \tau)$ be a tracial W^* -probability space and $x_1, \dots, x_d$ be a family of elements in $\mathcal{M}$ such that $x_i = x_{d-i+2}$ for $i = 2, 3, \dots, d$. An operator-valued $d \times d$ self-adjoint circulant matrix over $(x_1, \dots, x_d)$ is the matrix of the form

$$C_d(x_1, \dots, x_d) := \frac{1}{\sqrt{d}} \begin{pmatrix} x_1 & x_2 & \cdots & x_d \\ x_d & x_1 & \cdots & x_{d-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_2 & x_3 & \cdots & x_1 \end{pmatrix}.$$

We are interested in the case of $Nd \times Nd$ self-adjoint block matrices with circulant structure in which $x_1, \dots, x_d$ are independent and identically distributed $N \times N$ Wigner matrices. For this, let $(a_{ij})_{i,j \geq 1}$ be a family of independent and identically distributed random variables such that $\mathbb{E}(a_{ij}^2) = 0$, $\mathbb{E}(|a_{ij}|^2) = 1$ and $\mathbb{E}(|a_{ij}|^4) < \infty$. Recall the mapping in (4.1) and let A be the $N \times N$ Wigner matrix given by $A = A((a_{ij})_{1 \leq j \leq i \leq N})$. Let $A^{(1)}, \dots, A^{(\lfloor \frac{d}{2} \rfloor + 1)}$ be independent copies of the Wigner matrix A and then set $A^{(i)} = A^{(d-i+2)}$ for $i = \lfloor \frac{d}{2} \rfloor + 1, \dots, d$. Denote by X_N the $Nd \times Nd$ self-adjoint block circulant matrix given by $X_N := C_d(A^{(1)}, \dots, A^{(d)})$.

It has been shown in [25] that the limiting spectral distribution of X_N is a weighted sum of two semicircular distributions, i.e. has an SS -law as named by Girko in [11]. We provide now a quantitative version of this result:

Theorem 6.3. *For fixed $d \geq 1$, there exists $K > 0$ such that for any z with $\text{Im}(z) > K$ we have*

$$|\mathbb{E}[\text{tr}_{dN}(G_{X_N}(z))] - \tau(G_d(z))| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right),$$

where G_d is the Cauchy transform associated to the distribution μ_d given by

$$\mu_d = \begin{cases} \frac{d-1}{d}\gamma_{\frac{d-1}{d}} + \frac{1}{d}\gamma_{\frac{1}{d}} & \text{if } d \text{ is odd,} \\ \frac{d-2}{d}\gamma_{\frac{d-2}{d}} + \frac{2}{d}\gamma_{\frac{2}{d}} & \text{if } d \text{ is even} \end{cases}$$

and γ_{σ^2} denotes the centered semicircular distribution of variance σ^2 . Furthermore, the distribution of X_N converges to μ_d in Kolmogorov distance.

Proof. Let us first introduce some notation. Consider the family of $d \times d$ self-adjoint circular random matrices $\{A_{ij} : i, j \geq 1\}$ such that $A_{ij} = A_{ji}^*$ and $A_{ij} = C_d(a_{ij}^{(1)}, a_{ij}^{(2)}, \dots, a_{ij}^{(d)})$. It

has been shown in [25] that X_N has the same spectral distribution as the $Nd \times Nd$ block Wigner $Y_N = A(A_{11}, A_{12}, \dots, A_{NN})$ where again A is the linear mapping defined in (4.1) that allows constructing operator-valued Wigner matrices. Note that for any $1 \leq j \leq i \leq N$, the matrices A_{ij} are independent and identically distributed, hence exchangeable. Since $\mathbb{E}(|a_{ij}|^4) < \infty$, then by a simple application of the noncommutative Lindeberg method as in Theorem 6.1, we can replace the A_{ij} 's by matrices bounded in norm and having the same covariance structure. This will be omitted here but we can assume from now on that our matrices are bounded in norm and thus Y_N fits nicely in the setting of Theorem 4.3. Now, set $n = N(N+1)/2$ and let $\bar{A}_{ij} = A_{ij} - \mu$ where $\mu = \frac{1}{n} \sum_{1 \leq j \leq i \leq N} A_{ij}$ and consider the variance map

$$\eta_n : M_d(L^\infty) \rightarrow M_d(L^\infty), \quad \eta_n(b) = \frac{1}{n} \sum_{j \leq i} \bar{A}_{ij} b \bar{A}_{ij}.$$

Therefore, we get by Theorem 4.3 there exists $K > 0$ such that for any z with $\text{Im}(z) > K$,

$$|\mathbb{E}[\text{tr}_{dN}(G_{Y_N}(z))] - \tau(G_{S_N}(z))| = \mathcal{O}_z\left(\frac{1}{\sqrt{N}}\right),$$

where S_N is a centered semicircular over $M_d(L^\infty)$ of variance η_n . Again Lemma 4.13 guarantees that for any $z \in \mathbb{C}^+$

$$|\tau[G_{S_N}(z)] - \tau[G_S(z)]| = \mathcal{O}_z\left(\frac{1}{N}\right)$$

where S is a centered semicircular variable over $M_d(\mathbb{C})$ of covariance mapping

$$\eta : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C}), \quad \eta(B) = \mathbb{E}[A_{11} B A_{11}].$$

One can directly show that S has distribution μ_d or alternatively follow similar computations as in the proof of Proposition 1 in [25]. The convergence in Kolmogorov distance now follows from [32, Theorem A.2] since the cumulative distribution of a sum of semicirculars (SS -law) is Lipschitz. □

7. PROOF OF THEOREM 3.5

In the commutative setting, the limiting distribution of a Wigner type matrix, whose entries are exchangeable random variables, was shown in [6] to be semicircular. This was done via the Lindeberg method that allows approximating the spectral distribution of the matrix by that of a GOE, a Wigner matrix with i.i.d. Gaussian entries, and then concluding its limit. Our proof is inspired from its commutative analogue and is also based on the Lindeberg method and a Gaussian interpolation technique. However, it is not a straightforward extension as it has to deal with all the difficulties arising in the noncommutative realm that need astute manipulation of some terms when exchangeable elements are considered and freeness is dropped.

We proceed now to the proof and consider the shifted random elements $\bar{x}_1, \dots, \bar{x}_n, \bar{y}_1, \dots, \bar{y}_n$ given by $\bar{x}_i = x_i - \frac{1}{n} \sum_{k=1}^n x_k$ and $\bar{y}_i = y_i - \frac{1}{n} \sum_{k=1}^n y_k$ and set

$$\bar{\mathbf{x}} = \sum_{i=1}^n \bar{x}_i \otimes^* a_i \quad \text{and} \quad \bar{\mathbf{y}} = \sum_{i=1}^n \bar{y}_i \otimes^* a_i,$$

where we recall that, for any $x \in \mathcal{M}$ and $a \in \mathcal{N}$, $x \otimes^* a = x \otimes a + x^* \otimes a^*$. To prove the main result, we shall first reduce our study to the *exchangeable* elements $(\bar{x}_1, \dots, \bar{x}_n)$ and $(\bar{y}_1, \dots, \bar{y}_n)$ that are such that $\sum_{i=1}^n \bar{x}_i = \sum_{i=1}^n \bar{y}_i = 0$.

To do so, we first remark that for any $z \in \mathbb{C}^+$, we have by the resolvent identity

$$\begin{aligned} |\tau \otimes \varphi(G_{\mathbf{x}}(z)) - \tau \otimes \varphi(G_{\bar{\mathbf{x}}}(z))| &= |\tau \otimes \varphi(G_{\mathbf{x}}(z)(\mathbf{x} - \bar{\mathbf{x}})G_{G_{\bar{\mathbf{x}}}(z)}(z))| \\ &\leq \frac{1}{\operatorname{Im}(z)^2} \left\| \left(\frac{1}{n} \sum_{k=1}^n x_k \right) \otimes^* \left(\sum_{i=1}^n a_i \right) \right\|_{L^1} \\ &\leq 2 \frac{K_1}{\operatorname{Im}(z)^2} \left\| \frac{1}{n} \sum_{i=1}^n x_i \right\|_{\infty}. \end{aligned}$$

By exchangeability, we have

$$\left\| \frac{1}{n} \sum_{k=1}^n x_k \right\|_{\infty} \leq \frac{1}{n} \sum_{k=1}^n \|x_k\|_{\infty} = \|x_1\|_{\infty}.$$

Similarly, we get

$$|\mathbb{E}[\varphi(G_{\mathbf{y}}(z)) - \varphi(G_{\bar{\mathbf{y}}}(z))]| \leq 2 \frac{K_1}{\operatorname{Im}(z)^2} \mathbb{E} \left\| \frac{1}{n} \sum_{k=1}^n y_k \right\|_{\infty},$$

with

$$\frac{1}{n} \sum_{k=1}^n y_k = \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{\sqrt{n}} \sum_{\ell=1}^n N_{k,\ell} \right) x_{\ell} - \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{\sqrt{n}} \sum_{\ell=1}^n N_{k,\ell} \right) \left(\frac{1}{n} \sum_{j=1}^n x_j \right)$$

which can be bounded as follows:

$$\mathbb{E} \left\| \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{\sqrt{n}} \sum_{\ell=1}^n N_{k,\ell} \right) x_{\ell} \right\|_{\infty} \leq \frac{1}{n} \sum_{\ell=1}^n \mathbb{E} \left| \frac{1}{\sqrt{n}} \sum_{k=1}^n N_{k,\ell} \right| \|x_1\|_{\infty} \leq \|x_1\|_{\infty}$$

and

$$\mathbb{E} \left\| \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{\sqrt{n}} \sum_{\ell=1}^n N_{k,\ell} \right) \left(\frac{1}{n} \sum_{j=1}^n x_j \right) \right\|_{\infty} \leq \frac{1}{n} \sum_{\ell=1}^n \mathbb{E} \left| \frac{1}{\sqrt{n}} \sum_{k=1}^n N_{k,\ell} \right| \|x_1\|_{\infty} \leq \|x_1\|_{\infty}.$$

Thus, we get that

$$|\tau \otimes \varphi(G_{\mathbf{x}}(z)) - \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{y}}(z))]| \leq 6 \frac{K_1}{\operatorname{Im}(z)^2} \|x_1\|_{\infty} + |\tau \otimes \varphi(G_{\bar{\mathbf{x}}}(z)) - \mathbb{E}[\tau \otimes \varphi(G_{\bar{\mathbf{y}}}(z))]|.$$

It remains to control the term

$$|\tau \otimes \varphi(G_{\bar{\mathbf{x}}}(z)) - \mathbb{E}[\tau \otimes \varphi(G_{\bar{\mathbf{y}}}(z))]|.$$

This is done in the following Proposition 7.1 by which we can conclude our main statement.

Proposition 7.1. *Let $(\mathcal{M}, \tau)$ and $(\mathcal{N}, \varphi)$ be two tracial W^* -probability spaces. Let $(x_1, \dots, x_n)$ be an n -tuple of exchangeable elements in $\mathcal{M}$ such that $\sum_{i=1}^n x_i = 0$. Consider a family $(N_{i,k})_{1 \leq i, k \leq n}$ of independent standard Gaussian random variables and let $(y_1, \dots, y_n)$ be the n -tuple of random elements in $\mathcal{M}$ given by*

$$y_i = \frac{1}{\sqrt{n}} \sum_{k=1}^n N_{i,k} x_k$$

and consider the shifted random elements $(\bar{y}_1, \dots, \bar{y}_n)$ given by

$$\bar{y}_i = y_i - \frac{1}{n} \sum_{k=1}^n y_k.$$

Let $(a_1, \dots, a_n)$ be an n -tuple of elements in $\mathcal{N}$ and set

$$\mathbf{x} = \sum_{i=1}^n x_i \otimes a_i + x_i^* \otimes a_i^* \quad \text{and} \quad \bar{\mathbf{y}} = \sum_{i=1}^n \bar{y}_i \otimes a_i + \bar{y}_i^* \otimes a_i^*.$$

Then, for any $z \in \mathbb{C}^+$,

$$|\tau \otimes \varphi(G_{\mathbf{x}}(z)) - \mathbb{E}[\tau \otimes \varphi(G_{\bar{\mathbf{y}}}(z))]| \leq C \cdot \left(\frac{K_2^2}{\text{Im}(z)^3} \|x_1\|_\infty^2 \sqrt{n} + \frac{K_\infty K_2^2}{\text{Im}(z)^4} \|x_1\|_\infty^3 n + \frac{K_\infty^2 K_2^2}{\text{Im}(z)^5} \|x_1\|_\infty^4 n \right)$$

where $K_\infty = \max_i \|a_i\|_\infty$, $K_2 = \max_i \|a_i\|_{L^2}$ and C is a universal constant.

Proof. Consider $(u_1, \dots, u_n)$ given by

$$u_i = x_i + \frac{1}{n-i+1} \sum_{j<i} x_j = x_i - \frac{1}{n-i+1} \sum_{j \geq i} x_j,$$

and the random elements $(v_1, \dots, v_n)$ given by

$$v_i = \frac{1}{\sqrt{n}} \sum_{k=1}^n \tilde{N}_{i,k} x_k,$$

where $(\tilde{N}_{i,k})_{1 \leq i, k \leq n}$ is a family of independent standard Gaussian random variables which are independent from $(N_{i,k})_{1 \leq i, k \leq n}$. Let $(b_1, \dots, b_n)$ be given for each $i = 1, \dots, n$ by

$$b_i = a_i - \frac{1}{n-i} \sum_{i < j} a_j$$

in such a way that

$$\mathbf{x} = \sum_{i=1}^n x_i \otimes^* a_i = \sum_{i=1}^n u_i \otimes^* b_i.$$

Also we set $\mathbf{v} = \sum_{i=1}^n v_i \otimes^* b_i$. The proof will be done in two major steps: for any $z \in \mathbb{C}^+$, we first prove via the Lindeberg method in Theorem 3.1:

$$(7.1) \quad \begin{aligned} & |\tau \otimes \varphi(G_{\mathbf{x}}(z)) - \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{v}}(z))]| \\ & \leq C \left(\frac{K_2^2}{\text{Im}(z)^3} \|x_1\|_\infty^2 \sqrt{n} + \frac{K_\infty K_2^2}{\text{Im}(z)^4} \|x_1\|_\infty^3 n + \frac{K_\infty^2 K_2^2}{\text{Im}(z)^5} \|x_1\|_\infty^4 n \right); \end{aligned}$$

and then via a Gaussian interpolation technique:

$$(7.2) \quad \left| \mathbb{E}[\tau \otimes \varphi(G_{\bar{\mathbf{y}}}(z))] - \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{v}}(z))] \right| \leq \frac{3K_2^2}{\text{Im}(z)^3} \|x_1\|_\infty^2 \sqrt{n}.$$

Application of the Lindeberg method. Considering the n -tuples $(u_1, \dots, u_n)$ and $(v_1, \dots, v_n)$ in $\mathcal{M}$ and $(b_1, \dots, b_n)$ in $\mathcal{N}$, we apply Theorem 3.1 and get, for any $z \in \mathbb{C}^+$,

$$|\tau \otimes \varphi(G_{\mathbf{x}}(z)) - \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{v}}(z))]| \leq \sum_{i=1}^n (|\mathbb{E}[P_i]| + |\mathbb{E}[Q_i]| + |\mathbb{E}[R_i]|),$$

where in this case $\mathbf{z}_i$ and $\mathbf{z}_i^0$ are given by

$$\mathbf{z}_i = u_1 \otimes^* b_1 + \dots + u_i \otimes^* b_i + v_{i+1} \otimes^* b_{i+1} + \dots + v_n \otimes^* b_n,$$

$$\mathbf{z}_i^0 = u_1 \otimes^* b_1 + \dots + u_{i-1} \otimes^* b_{i-1} + v_{i+1} \otimes^* b_{i+1} + \dots + v_n \otimes^* b_n,$$

and the terms P_i, Q_i and R_i are thus:

$$P_i = \tau \otimes \varphi[G_{\mathbf{z}_i^0}(z)(u_i \otimes^* b_i - v_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)],$$

$$Q_i = \tau \otimes \varphi[G_{\mathbf{z}_i^0}(z)(u_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(u_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)] - \tau \otimes \varphi[G_{\mathbf{z}_i^0}(z)(v_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(v_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)],$$

and

$$R_i = \tau \otimes \varphi[G_{\mathbf{z}_i}(z)((u_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z))^3] - \tau \otimes \varphi[G_{\mathbf{z}_{i-1}}(z)((v_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z))^3].$$

Now to control the above quantities, it is convenient to introduce first some notation: for all $i, j, k \in \{1, \dots, n\}$, let $u_i^{j \leftrightarrow k}$ and $v_i^{j \leftrightarrow k}$ denote the elements defined exactly as u_i and v_i but by swapping x_j and x_k . We also set

$$\mathbf{z}_i^{0, j \leftrightarrow k} = u_1^{j \leftrightarrow k} \otimes^* b_1 + \dots + u_{i-1}^{j \leftrightarrow k} \otimes^* b_{i-1} + v_{i+1}^{j \leftrightarrow k} \otimes^* b_{i+1} + \dots + v_n^{j \leftrightarrow k} \otimes^* b_n.$$

A key point of the proof is the following lemma which follows from the exchangeability of the x_i 's.

Lemma 7.2. *Let $i \leq j$ and $i \leq k$ then for any polynomial P in three noncommuting variables, we have*

$$\mathbb{E}[\tau \otimes \varphi[P(G_{\mathbf{z}_i^0}(z), x_j \otimes^* b_i, x_k \otimes^* b_i)]] = \mathbb{E}[\tau \otimes \varphi[P(G_{\mathbf{z}_i^0}(z), x_k \otimes^* b_i, x_j \otimes^* b_i)]].$$

Proof of the Lemma 7.2. Note that $(v_1, \dots, v_n)$ has the same law as $(v_1^{j \leftrightarrow k}, \dots, v_n^{j \leftrightarrow k})$ as a random variable, and thus, if $i \leq j$ and $i \leq k$, then $\mathbf{z}_i^{0, j \leftrightarrow k}$ has the same law as $\mathbf{z}_i^0$ as a random variable. As a consequence, we just need to prove that

$$\tau \otimes \varphi[P(G_{\mathbf{z}_i^0}(z), x_j \otimes^* b_i, x_k \otimes^* b_i)] = \tau \otimes \varphi[P(G_{\mathbf{z}_i^{0, j \leftrightarrow k}}(z), x_k \otimes^* b_i, x_j \otimes^* b_i)].$$

This equality, valid for all $i, j, k \in \{1, \dots, n\}$, is a direct consequence of exchangeability. Indeed, for any polynomial Q , we have

$$\tau[Q(u_1, \dots, u_{i-1}, v_{i+1}, \dots, v_n, x_j, x_k)] = \tau[Q(u_1, \dots, u_{i-1}, v_{i+1}^{j \leftrightarrow k}, \dots, v_n^{j \leftrightarrow k}, x_k, x_j)],$$

from which we deduce that, for any polynomial Q , we have

$$\begin{aligned} & \tau \otimes \varphi[Q(u_1 \otimes^* b_1, \dots, u_{i-1} \otimes^* b_{i-1}, v_{i+1} \otimes^* b_{i+1}, \dots, v_n \otimes^* b_n, x_j \otimes^* b_i, x_k \otimes^* b_i)] \\ &= \tau \otimes \varphi[Q(u_1 \otimes^* b_1, \dots, u_{i-1} \otimes^* b_{i-1}, v_{i+1}^{j \leftrightarrow k} \otimes^* b_{i+1}, \dots, v_n^{j \leftrightarrow k} \otimes^* b_n, x_k \otimes^* b_i, x_j \otimes^* b_i)]. \end{aligned}$$

Finally, by an approximation procedure, we get

$$\tau \otimes \varphi[P(G_{\mathbf{z}_i^0}(z), x_j \otimes^* b_i, x_k \otimes^* b_i)] = \tau \otimes \varphi[P(G_{\mathbf{z}_i^{0, j \leftrightarrow k}}(z), x_k \otimes^* b_i, x_j \otimes^* b_i)].$$

□

By Lemma 7.2, we have for $j \geq i$,

$$\mathbb{E}\left[\tau \otimes \varphi\left[G_{\mathbf{z}_i^0}(z)(x_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right]\right] = \mathbb{E}\left[\tau \otimes \varphi\left[G_{\mathbf{z}_i^0}(z)(x_j \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right]\right]$$

which implies that

$$\mathbb{E}\left[\tau \otimes \varphi\left[G_{\mathbf{z}_i^0}(z)(u_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right]\right] = 0.$$

On the other hand, we note that $\mathbb{E}[v_i \otimes^* b_i] = 0$ and $v_i \otimes^* b_i$ is independent from $\mathbf{z}_i^0$, which imply that

$$\mathbb{E}\left[\tau \otimes \varphi\left[G_{\mathbf{z}_i^0}(z)(v_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right]\right] = 0.$$

As a consequence, $\mathbb{E}[P_i] = 0$. Now turning to Q_i and again by using Lemma 7.2, we get for any $\ell \geq i$

$$\begin{aligned} & \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(x_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i\right)G_{\mathbf{z}_i^0}(z)\right)\right] \\ &= \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(x_\ell \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i\right)G_{\mathbf{z}_i^0}(z)\right)\right] \\ &= \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)\left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i\right)G_{\mathbf{z}_i^0}(z)\left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i\right)G_{\mathbf{z}_i^0}(z)\right)\right] \end{aligned}$$

and thus we get

$$\begin{aligned} & \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(u_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(u_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right)\right] \\ &= \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(x_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(x_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right)\right] \\ & \quad - \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)\left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i\right)G_{\mathbf{z}_i^0}(z)\left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i\right)G_{\mathbf{z}_i^0}(z)\right)\right]. \end{aligned}$$

On the other hand, we have

$$\begin{aligned} & \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(v_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(v_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right)\right] \\ &= \frac{1}{n} \sum_{k, \ell} \mathbb{E}(\tilde{N}_{i,k} \tilde{N}_{i,\ell}) \mathbb{E}\left[\tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(x_k \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(x_\ell \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right)\right] \\ &= \mathbb{E}\left[\frac{1}{n} \sum_k \tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(x_k \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(x_k \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right)\right] \\ &= \mathbb{E}\left[\frac{1}{n} \sum_{k < i} \tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(x_k \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(x_k \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right)\right] \\ & \quad + \mathbb{E}\left[\frac{n-i+1}{n} \tau \otimes \varphi\left(G_{\mathbf{z}_i^0}(z)(x_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)(x_i \otimes^* b_i)G_{\mathbf{z}_i^0}(z)\right)\right]. \end{aligned}$$

As a consequence, $|\mathbb{E}[Q_i]|$ is bounded by the terms

$$\begin{aligned} & \left| \mathbb{E} \left[\frac{i-1}{n} \tau \otimes \varphi \left(G_{\mathbf{z}_i^0}(z) (x_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right) \right] \right. \\ & \quad \left. - \mathbb{E} \left[\frac{1}{n} \sum_{k < i} \tau \otimes \varphi \left(G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right) \right] \right| \\ & + \left| \mathbb{E} \left[\tau \otimes \varphi \left(G_{\mathbf{z}_i^0}(z) \left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i \right) G_{\mathbf{z}_i^0}(z) \left(\frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i \right) G_{\mathbf{z}_i^0}(z) \right) \right] \right| \\ & := |\mathbb{E}[Q_i^{(1)}]| + |\mathbb{E}[Q_i^{(2)}]|. \end{aligned}$$

As in the demonstration of Lemma 7.2, one can prove for any $k < i$,

$$\begin{aligned} \tau \otimes \varphi [G_{\mathbf{z}_i^0}(z) (x_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z)] \\ = \tau \otimes \varphi [G_{\mathbf{z}_i^{0,i \leftrightarrow k}}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^{0,i \leftrightarrow k}}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^{0,i \leftrightarrow k}}(z)]. \end{aligned}$$

Using the resolvent identity in Lemma 3.3 with $m = 1$, we compute for any $k < i$,

$$\begin{aligned} & \left| \tau \otimes \varphi [G_{\mathbf{z}_i^{0,i \leftrightarrow k}}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^{0,i \leftrightarrow k}}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^{0,i \leftrightarrow k}}(z)] \right. \\ & \quad \left. - \tau \otimes \varphi [G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z)] \right| \\ & \leq \frac{3}{\text{Im}(z)^2} \|G_{\mathbf{z}_i^{0,i \leftrightarrow k}}(z) - G_{\mathbf{z}_i^0}(z)\|_\infty \|x_k \otimes^* b_i\|_{L^2}^2 \\ & \leq \frac{48}{\text{Im}(z)^4} \|\mathbf{z}_i^{0,i \leftrightarrow k} - \mathbf{z}_i^0\|_\infty K_2^2 \|x_1\|_\infty^2. \end{aligned}$$

Note that, for all $j \in \{1, \dots, n\}$, $\|u_j\|_\infty \leq 2\|x_1\|_\infty$ and $\|b_j\|_\infty \leq 2K_\infty$. As a consequence,

$$\begin{aligned} \|\mathbf{z}_i^{0,i \leftrightarrow k} - \mathbf{z}_i^0\|_\infty &= \|u_k \otimes^* b_i + u_i \otimes^* b_k - u_k \otimes^* b_k - u_i \otimes^* b_i\|_\infty \\ &\leq 32\|x_1\|_\infty K_\infty \end{aligned}$$

and thus

$$|\mathbb{E}[Q_i^{(1)}]| \leq C \frac{K_\infty K_2^2}{\text{Im}(z)^4} \|x_1\|_\infty^3$$

whenever $C \geq 1536$. On the other hand, turning to the second term $\mathbb{E}[Q_i^{(2)}]$, we get

$$|\mathbb{E}[Q_i^{(2)}]| \leq \frac{1}{\text{Im}(z)^3} \left\| \frac{1}{n-i+1} \sum_{j \geq i} x_j \otimes^* b_i \right\|_{L^2}^2 \leq 16 \frac{K_2^2}{\text{Im}(z)^3} \left\| \frac{1}{n-i+1} \sum_{j \geq i} x_j \right\|_{L^2}^2.$$

Using exchangeability and the same computation as in Chatterjee [6, eq. (4)], we get

$$\left\| \frac{1}{n-i+1} \sum_{j \geq i} x_j \right\|_{L^2}^2 = \frac{i-1}{(n-i+1)(n-1)} \|x_1\|_{L^2}^2 \leq \frac{1}{n-i+1} \|x_1\|_{L^2}^2.$$

Collecting all the above bounds, we finally get

$$|\mathbb{E}[Q_i]| \leq C \frac{K_2^2}{\text{Im}(z)^3} \frac{1}{n-i+1} \|x_1\|_{L^2}^2 + C \frac{K_\infty K_2^2}{\text{Im}(z)^4} \|x_1\|_\infty^3$$

where C is a sufficiently large constant. We control now the third order term R_i . Considering its first term, we get the following bound:

$$\begin{aligned} \left| \mathbb{E} \left[\tau \otimes \varphi \left[G_{\mathbf{z}_i}(z) \left((u_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right)^3 \right] \right] \right| &\leq \frac{1}{Im(z)^4} \|u_i \otimes^* b_i\|_\infty \|u_i \otimes^* b_i\|_{L^2}^2 \\ &\leq 64 \frac{K_\infty K_2^2}{Im(z)^4} \|u_i\|_\infty \|u_i\|_{L^2}^2 \\ &\leq 512 \frac{K_\infty K_2^2}{Im(z)^4} \|x_1\|_\infty^3. \end{aligned}$$

Recall the second term in R_i :

$$\tau \otimes \varphi \left[G_{\mathbf{z}_{i-1}}(z) \left((v_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right)^3 \right].$$

Developing the last $v_i = \frac{1}{\sqrt{n}} \sum_k \tilde{N}_{i,k} x_k$ in it, we get n terms of the form

$$\begin{aligned} &\frac{1}{\sqrt{n}} \left| \mathbb{E} \left[\tilde{N}_{i,k} \tau \otimes \varphi \left[G_{\mathbf{z}_{i-1}}(z) \left((v_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right)^2 (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right] \right] \right| \\ &= \frac{1}{n} \left| \mathbb{E} \left[\tau \otimes \varphi \left[G_{\mathbf{z}_{i-1}}(z) (v_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right] \right] \right| \\ &\quad + \frac{1}{n} \left| \mathbb{E} \left[\tau \otimes \varphi \left[G_{\mathbf{z}_{i-1}}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (v_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right] \right] \right| \\ &\quad + \frac{1}{n} \left| \mathbb{E} \left[\tau \otimes \varphi \left[G_{\mathbf{z}_{i-1}}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_{i-1}}(z) (v_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (v_i \otimes^* b_i) G_{\mathbf{z}_i^0}(z) (x_k \otimes^* b_i) G_{\mathbf{z}_i^0}(z) \right] \right] \right| \end{aligned}$$

where the above equality is obtained via an integration by parts. The third term is bounded by

$$\frac{1}{n} \frac{1}{Im(z)^5} \|x_k \otimes^* b_i\|_\infty^2 \mathbb{E} \|v_i \otimes^* b_i\|_{L^2}^2 \leq C \frac{1}{n} \frac{K_\infty^2 K_2^2}{Im(z)^5} \|x_1\|_\infty^2 \mathbb{E} \|v_i\|_{L^2}^2 \leq C \frac{1}{n} \frac{K_\infty^2 K_2^2}{Im(z)^5} \|x_1\|_\infty^4$$

since

$$\mathbb{E} \|v_i\|_{L^2}^2 = \frac{1}{n} \sum_{k,l} \mathbb{E} [\tilde{N}_{i,k} \tilde{N}_{i,l}] \tau(x_k x_l) = \frac{1}{n} \sum_k \|x_1\|_{L^2}^2 = \|x_1\|_{L^2}^2.$$

Similarly, the first and second terms are bounded by

$$C \frac{1}{n} \frac{K_\infty K_2^2}{Im(z)^4} \|x_1\|_\infty^3.$$

Collecting the above bounds we finally get,

$$|\mathbb{E}[R_i]| \leq C \frac{K_\infty K_2^2}{Im(z)^4} \|x_1\|_\infty^3 + C \frac{K_\infty^2 K_2^2}{Im(z)^5} \|x_1\|_\infty^4.$$

Noting that $\sum_{i=1}^n \frac{1}{n-i+1} \leq C\sqrt{n}$ whenever C is sufficiently large, then gathering the bounds on Q_i and R_i and summing over $i = 1, \dots, n$, we end the proof of (7.1).

Gaussian Interpolation. To prove (7.2), we set for each $0 \leq t \leq 1$,

$$\mathbf{w}_t = \sqrt{1-t}\mathbf{v} + \sqrt{t}\bar{\mathbf{y}}.$$

Then

$$\begin{aligned} \mathbb{E}[\tau \otimes \varphi[G_{\bar{\mathbf{y}}}(z)]] - \mathbb{E}[\tau \otimes \varphi[G_{\mathbf{v}}(z)]] &= \mathbb{E} \int_0^1 \partial_t \tau \otimes \varphi[G_{\mathbf{w}_t}(z)] dt \\ &= \mathbb{E} \int_0^1 \tau \otimes \varphi \left[G_{\mathbf{w}_t}(z)^2 \left(\frac{\bar{\mathbf{y}}}{2\sqrt{t}} - \frac{\mathbf{v}}{2\sqrt{1-t}} \right) \right] dt. \end{aligned}$$

We can decompose by linearity

$$\bar{\mathbf{y}} = \frac{1}{\sqrt{n}} \sum_{i,k=1}^n \left(N_{i,k} - \frac{1}{n} \sum_{j=1}^n N_{j,k} \right) x_k \otimes^* a_i := \frac{1}{\sqrt{n}} \sum_{i,k=1}^n W_{i,k} x_k \otimes^* a_i$$

and

$$\mathbf{v} = \frac{1}{\sqrt{n}} \sum_{i,k=1}^n \left(\tilde{N}_{i,k} - \sum_{j=1}^{i-1} \frac{1}{n-j} \tilde{N}_{j,k} \right) x_k \otimes^* a_i := \frac{1}{\sqrt{n}} \sum_{i,k=1}^n \tilde{W}_{i,k} x_k \otimes^* a_i.$$

By integration by parts, we get

$$\begin{aligned} \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{w}_t}(z)^2 \bar{\mathbf{y}})] &= \frac{1}{\sqrt{n}} \sum_{i,k=1}^n \mathbb{E}[W_{i,k} \tau \otimes \varphi(G_{\mathbf{w}_t}(z)^2 (x_k \otimes^* a_i))] \\ &= 2\sqrt{t} \sum_{i,i'=1}^n \sigma_{i,i'} \frac{1}{n} \sum_{k=1}^n \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{w}_t}(z)(x_k \otimes^* a_{i'}) G_{\mathbf{w}_t}(z)^2 (x_k \otimes^* a_i))], \end{aligned}$$

where for any $k \in \{1, \dots, n\}$,

$$\sigma_{i,i'} := \mathbb{E}[W_{i,k} W_{i',k}] = \begin{cases} -n^{-1} & \text{if } i \neq i' \\ (n-1)/n & \text{if } i = i'. \end{cases}$$

Similarly, we get

$$\begin{aligned} \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{w}_t}(z)^2 \mathbf{v})] &= \frac{1}{\sqrt{n}} \sum_{i,k=1}^n \mathbb{E}[\tilde{W}_{i,k} \tau \otimes \varphi(G_{\mathbf{w}_t}(z)^2 (x_k \otimes^* a_i))] \\ &= 2\sqrt{1-t} \sum_{i,i'=1}^n \tilde{\sigma}_{i,i'} \frac{1}{n} \sum_{k=1}^n \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{w}_t}(z)(x_k \otimes^* a_{i'}) G_{\mathbf{w}_t}(z)^2 (x_k \otimes^* a_i))], \end{aligned}$$

where for any $k \in \{1, \dots, n\}$,

$$\tilde{\sigma}_{i,i'} := \mathbb{E}[\tilde{W}_{i,k} \tilde{W}_{i',k}] = \begin{cases} -\frac{1}{n-1} - \sum_{j=1}^{i-1} \frac{1}{(n-j)^2(n-j-1)} & \text{if } i < i' \\ 1 + \sum_{j=1}^{i-1} \frac{1}{(n-j)^2} & \text{if } i = i' \\ \tilde{\sigma}_{i',i} & \text{if } i > i'. \end{cases}$$

Combining the above quantities, we get

$$\begin{aligned}
& \left| \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{y}}(z))] - \mathbb{E}[\tau \otimes \varphi(G_{\mathbf{v}}(z))] \right| \\
&= \sum_{i,i'=1}^n |\sigma_{i,i'} - \tilde{\sigma}_{i,i'}| \frac{1}{n} \sum_{k=1}^n \int_0^1 \mathbb{E} |\tau \otimes \varphi(G_{\mathbf{w}_t}(z)(x_k \otimes^* a_{i'}) G_{\mathbf{w}_t}(z)^2(x_k \otimes^* a_i))| dt \\
&\leq 4 \sum_{i,i'=1}^n |\sigma_{i,i'} - \tilde{\sigma}_{i,i'}| \frac{K_2^2}{Im(z)^3} \|x_1\|_\infty^2 \\
&\leq 12 \frac{K_2^2}{Im(z)^3} \|x_1\|_\infty^2 \sqrt{n},
\end{aligned}$$

where the last inequality follows by the same computation as in Chatterjee [6, eq. (12)] to show that

$$\sum_{i,i'=1}^n |\sigma_{i,i'} - \tilde{\sigma}_{i,i'}| \leq 3 + \sum_{k=2}^{n-1} \frac{1}{k} \leq 3\sqrt{n}.$$

□

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